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Pore-scale evaluation of hydrogen storage and recovery in basaltic formations

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ABSTRACT

Underground hydrogen storage (UHS) in basaltic rocks offers a scalable solution for large-scale sustainable energy needs, yet its efficiency is limited by poorly constrained pore-scale hysteresis during cyclic hydrogen-brine flow. While basaltic rocks have been extensively studied for carbon sequestration and critical mineral extraction, the pore-scale physics governing cyclic hydrogen-brine interactions, particularly the roles of snap-off, wettability, and hysteresis, remain inadequately understood. This knowledge gap hinders the development of predictive models and optimization strategies for UHS performance.

This study presents a pore-scale investigations of cyclic hydrogen-brine flow in basaltic formations, combining micro-computed tomography imaging with pore network modelling. A systematic workflow is employed to evaluate the effects of successive drainage-imbibition cycles on multiphase flow properties under varying wetting regimes, with emphasis on hysteresis evolution and its influence on recoverable hydrogen. Model validation is achieved through a novel benchmarking approach that incorporates synthetic fractures and morphological scaling, enabling calibration against experimental capillary pressure and relative permeability.

Results show that hydrogen trapping is primarily governed by snap-off and pore-body isolation, particularly within large, angular pores exhibiting high aspect ratios and limited connectivity. Strong hysteresis is observed between drainage and imbibition, with hydrogen saturations averaging 85%, predominantly in larger pore spaces, compared to a residual saturation of 61% following imbibition. Repeated cycling leads to a gradual increase in residual saturation, which eventually stabilizes, indicating the onset of a hysteresis equilibrium state. Wettability emerges as a critical second-order control on displacement dynamics. Shifting from strongly to weakly water-wet conditions reduces capillary entry pressures, enhances brine re-invasion, and increases hydrogen recovery efficiency by ∼6%. These findings offer mechanistic insights into capillary trapping and wettability effects, providing a framework for optimizing UHS in reactive and abundant, yet underutilized, basalt formations, and supporting ongoing global decarbonization efforts through reliable subsurface hydrogen storage.

1. Introduction

The global push towards a sustainable energy future and the urgent need to mitigate climate change have intensified interest in hydrogen, a versatile energy carrier producible through various pathways, often classified by colour designations such as blue, green, white, and orange hydrogen (Osselin et al., 2022). Hydrogen is particularly attractive for its potential to reduce reliance on non-renewable energy sources while decarbonizing key sectors, including transportation, heavy industry, and long-term energy storage (Pan et al., 2021; Reitenbach et al., 2015). However, the scalability and the reliability of hydrogen deployment hinge not only on its production technologies but also on the development of effective, large-scale storage solutions (Heinemann et al., 2021). The unique properties of hydrogen, especially its low density and high diffusivity, pose significant constraints on storing it in vast quantities using conventional means (Hosseini et al., 2022; Krevor et al., 2023). While surface-based hydrogen storage facilities and processes (like chemical storage, compression, and liquefaction) exist and continue to be developed to achieve useful energy density, they remain cost-intensive and limited in capacity (typically supporting discharge durations of only hours to days), making them insufficient for grid-level applications that must accommodate intermittent nature of renewable energy sources and the associated mismatch between energy supply and demand (Goodarzi et al., 2024; Fekete et al., 2023; Summers et al., 2024). As a result, underground hydrogen storage (UHS) in geologic formations, such as depleted gas reservoirs, saline aquifers,

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and salt caverns has emerged as a promising large-scale option, offering gigawatt- to terawatt-hour storage capacities that far exceed those of surface-based systems (Wang et al., 2023b; Uliasz-Misiak et al., 2025; Gomez Mendez et al., 2024).

However, much of the current conceptual and operational framework for UHS has been adapted from established practices in natural gas storage and geological CO_2 sequestration (Heinemann et al., 2021). While these analogs provide preliminary evaluations of UHS feasibility, key differences in the physicochemical and hydrodynamic characteristics of hydrogen (H_2) relative to other gases necessitate a re-examination of existing frameworks. For instance, interfacial properties governing fluid behaviour and distribution within porous media in CO_2-fluid systems differ from those observed in H_2-fluid systems (Kamari et al., 2017; Mouallem et al., 2024). Moreover, unlike CO_2 injection, which is intended to remain permanently trapped, UHS requires efficient recovery of stored H_2 to satisfy variable energy demand (Tarkowski and Uliasz-Misiak, 2022; Zivar et al., 2021). Despite its promise, several scientific and technical challenges continue to limit the large-scale deployment of UHS. These include hydrogen losses due to microbial activity (Dopffel et al., 2021), geochemical reactions that may compromise reservoir integrity or alter fluid composition (Al-Yaseri et al., 2023; Zhan et al., 2024), and concerns over containment security and potential leakage pathways (Iglauer, 2022). Additional complexities arise from hydrogen mixing with residual or cushion gases (Krevor et al., 2023), salt precipitation and dry-out near injection wells during cycling (Bijay et al., 2024), and pronounced hysteresis in two-phase flow behaviour, which affects both injectivity and recovery efficiency (Higgs et al., 2024; Lysyy et al., 2022; Pan et al., 2023; Zhang et al., 2024).

Operationally, UHS systems function cyclically, alternating between H_2 injection and withdrawal in response to energy demand and the characteristics of the storage formation (Heinemann et al., 2021; Bijay et al., 2024). During injection (drainage), H_2 displaces the resident brine through immiscible displacement dynamics, preferentially occupying larger pores as the non-wetting phase, while pushing brine into smaller pores and interconnected throats (Boon et al., 2024a,b; Lysyy et al., 2022; Wang et al., 2023b, 2024). During withdrawal (imbibition), the displaced brine re-infiltrates the pores, displacing H_2 and trapping a portion of it as discrete droplets due to capillary forces, leading to droplet snap-off events and residual gas entrapment (Al-Yaseri et al., 2023). Over successive injection and withdrawal cycles, some disconnected H_2 bubbles may coalesce and regain mobility, causing fluid saturation to depend not only on the current drainage or imbibition process but also on the history of previous flow cycles, commonly referred as *hysteresis* (Li and Yao, 2023). Neglecting hysteresis can lead to underestimation of residual trapping and overestimation of extractable H_2, with observed reductions in H_2 recovery efficiency ranging from 16% to 25% (Bahrami et al., 2023; Ershadnia et al., 2023). This complex, hysteretic nature of immiscible flow highlights the need for further research to characterize how cyclic H_2-brine interactions impact residual saturation beyond primary drainage and imbibition events. Some studies, such as Shiran et al. (2024), report increasing residual H_2 saturation over successive cycles with no plateau observed (from ~0.05 in the 1st cycle to ~0.14 in the 3rd cycle), suggesting cumulative trapping effects. In contrast, others observe relatively consistent residual H_2 saturation after each imbibition cycle (Thaysen et al., 2023; Lysyy et al., 2023), pointing to a lack of consensus. This inconsistency underscores the importance of systematic investigations across multiple injection–withdrawal cycles to determine whether repeated cycling amplifies hysteresis and increases H_2 residual trapping, or reaches a stabilized equilibrium over time.

Most existing subsurface energy storage studies have focused primarily on sedimentary rocks, leaving a notable gap in the evaluation of alternative lithologies such as mafic and ultramafic rocks. Recent studies have highlighted the geochemical reactivity of these rocks and their dual utility in CO_2 sequestration and critical mineral recovery (Matter et al., 2016; Wang and Dreisinger, 2022; DePaolo et al., 2021; Awolayo et al., 2022; Stanfield et al., 2024). Their capacity to facilitate permanent carbon storage through mineralization, and to mobilize energy-relevant trace metals positions them as increasingly attractive for integrated resource and storage applications (Wang and Dreisinger, 2022; Stanfield et al., 2024). However, their potential and viability for UHS remains largely unexplored. Basalts, which constitutes ~8% of the Earth's continental crust and occur extensively in coastal and island settings, where onshore access and thick flow sequences provide both logistical and geological advantages (Dehghani et al., 2024; Stanfield et al., 2024; Snæbjörnsdóttir et al., 2020), present a compelling yet underexplored opportunity for UHS. Their high fracture connectivity, heterogeneous textures, and moderate subsurface temperatures are favourable for both fluid injectivity and storage capacity (Goldberg et al., 2008; Snæbjörnsdóttir et al., 2020).

Yet, their distinct mineralogy, complex pore structure, and potentially dynamic surface wettability pose unique uncertainties in hydrogen flow behaviour, capillary trapping, and recovery efficiency compared to conventional sedimentary settings (Stanfield et al., 2024; Al-Yaseri et al., 2023; Aftab et al., 2024). Moreover, recent findings suggest that microbially mediated changes in rock wettability through biofilm impregnation and surface chemistry changes during UHS can influence hydrogen phase distribution, retention and recovery (Ali et al., 2023; Boon et al., 2024b; Liu et al., 2023). Aftab et al. (2024) demonstrated that microbial colonization in a basalt-water-H_2 system shifted contact angles from strongly water-wet (12.7 ° and 19.5 °) to significantly less water-wet conditions (62 ° and 70 °), with implications for capillary trapping and gas mobility across injection-withdrawal cycles. These findings underscore the need for tailored pore-scale studies that capture the detailed displacement physics and integrate flow and wettability conditions, which are essential for evaluating the feasibility of UHS in reactive lithologies such as basalt.

This study addresses these critical knowledge gaps by leveraging a pore network modelling (PNM) framework to investigate cyclic H_2-brine interactions in a three-dimensional (3D) pore network reconstructed from X-ray microcomputed tomography (XCT) scans of basalt. We integrate XCT-based pore structure characterization and multiphase flow simulations to determine not only how much H_2 basalt can store, but also how much can be efficiently recovered. We quantify the evolution of hysteresis and evaluate how residual hydrogen trapping evolves over multiple injection-withdrawal cycles, whether cumulative or stabilizing, and how this behaviour is modulated under different wetting conditions (both pristine and microbially altered states) throughout successive cycles. This work represents the first systematic investigation of cyclic hysteresis and wettability effects on hydrogen storage efficiency and residual trapping in basaltic formations, advancing the scientific basis for expanding UHS into non-traditional lithologies.

2. Material and methods

2.1. Rock sample characterization

The rock sample used in this study was acquired from the Hawaii Scientific Drilling Project (DePaolo et al., 1996; Stolper et al., 2009). The site has been previously investigated for its carbon mineralization and critical mineral resource potential, as documented by DePaolo et al. (2021) and Stanfield et al. (2024). The sample was scanned using the Nikon M2 225 KV XCT at McMaster University's Canadian Centre for Electron Microscopy (CCEM), operating at an accelerating voltage of 140 kV and a target current of 150 μA. During the scanning process, the sample was continuously rotated within the X-ray beam, and a predetermined number of images were captured at each degree of rotation. These high-fidelity two-dimensional cross-sectional images captured at a voxel resolution of 8.57 μm were subsequently processed using Dragonfly image processing software (Comet Technologies Canada Inc.,

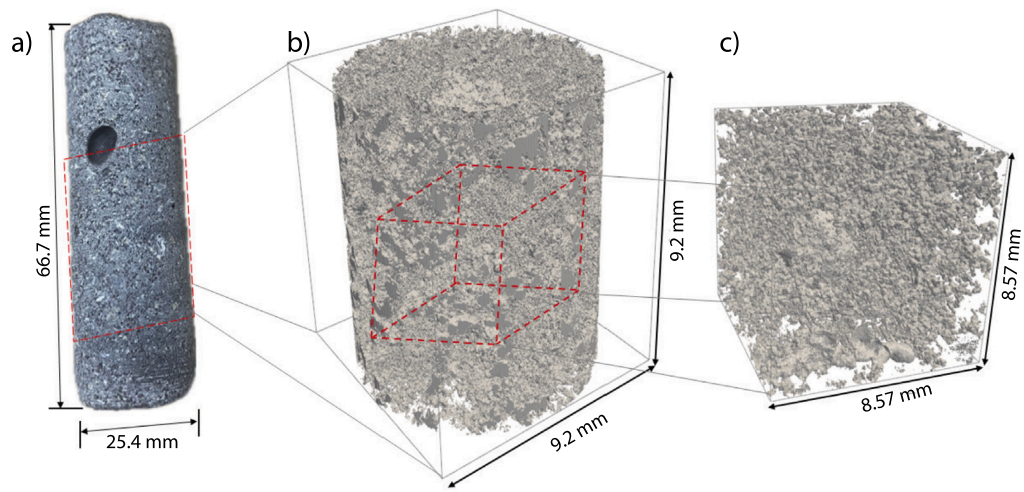

Fig. 1. Hawaiian basalt rock sample and subdomains used for image-based analysis. (a) the full core sample (66.7 mm × 25.4 mm) (b) a representative cylindrical volume (9.2 mm × 9.2 mm) was extracted from the central area as the region of interest (ROI). (c) cubic subdomain (8.57 mm × 8.57 mm × 8.57 mm) was selected from the ROI volume for subsequent detailed pore network extraction and pore-scale characterization.

2022), which applied image correction techniques and real-time volume rendering to construct a detailed 3D digital representation of the sample's internal structure (Fig. 1). Image segmentation and thresholding were performed using ImageJ Fiji^{TM} to create a binary image of solid and pore phases. The workflow used to carry out image processing and characterization is further described in the Supplementary Text.

Due to limitations in computational resources and the presence of imaging artifacts in the outer regions of the core (Fig. 1a), only a sub-volume of the basalt sample could be reconstructed at sufficient resolution. This sub-volume (Fig. 1b), referred to as the region of interest (ROI), was subsequently used for representative elementary volume (REV) analysis. The purpose was to identify the smallest subvolume over which porosity remains statistically invariant, thereby ensuring representativeness while maximizing computational efficiency. Subsequently, REV analysis was conducted on the reconstructed 3D digital image in Fig. 1b using the PoreSpy toolkit (Gostick et al., 2016; Wang et al., 2025). The full ROI was subdivided into 1000 cubic subdomains, each defined by a randomly assigned center coordinate and side lengths uniformly sampled between 10 and 1000 voxels. The upper bound corresponds to the smallest image dimension at which the subdomain porosity converged with that of the entire ROI. Boundary padding was incorporated to mitigate edge effects and ensure accurate porosity estimation. The local porosity, ϕ_i, of each subdomain was computed as the ratio of pore voxels to the total number of voxels within the respective volume, as:

$$\phi_i = \frac{N_{\text{pore},i}}{N_{\text{total},i}} \quad \text{where} \quad N_{\text{pore},i} = \sum_{j=1}^{N_{\text{total},i}} I_{\text{pore}}(\mathbf{x}_j)$$
$$\text{and} \quad I_{\text{pore}}(\mathbf{x}_j) = \begin{cases} 1, & \text{if voxel } j \text{ is pore,} \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

where $N_{\text{total},i}$ is the total voxel count in subdomain i. A detailed description of the REV procedure and associated parameters is provided in the Supplementary Text.

2.2. Network extraction and pore scale modelling

Pore network extraction was performed on the cubic REV digital image (Fig. 1c) using the maximal ball algorithm, which is well suited for preserving topological features and capturing complex porous architectures. This algorithm, extensively discussed by Dong and Blunt (2009) and Raeini et al. (2019), identifies pores and throats by computing the Euclidean distance map of the segmented pore space, allowing for the delineation of dilation and constriction regions. The size of each element was determined by fitting the largest possible inscribed spheres at each location (Fig. 2a), with their geometries classified as circular, triangular, or square cross sections based on their respective shape factors (Valvatne and Blunt, 2004; Hashemi et al., 2021; Hashemi and Vuik, 2024). Following the pore network extraction, a quasi-static pore network model (PNM) was employed to simulate multiphase flow, with key simulation parameters summarized in Table 1. This approach, which has been well demonstrated to reliably capture steady-state drainage and imbibition processes under controlled fluid configurations and wetting regimes (Cui et al., 2022; Hakimov et al., 2022; Rasmusson et al., 2018; Wang et al., 2021), assumes capillary-dominated flow (i.e., low capillary number – Ca), where fluid displacement is primarily governed by capillary and interfacial forces (Hashemi et al., 2021).

The quasi-static flow regime is characterized by a range of pore-scale displacement mechanisms, as illustrated in Fig. 2b-d. Notably, capillary trapping arises from these distinct mechanisms, including frontal (piston-like) displacement, snap-off driven by wetting layer swelling, and cooperative pore-body filling. The combination of these mechanisms determine whether H_2 moves freely or becomes immobilized within the pore network. Snap-off and in-type filling events (e.g., I_1, I_2), for example, are more likely in geometries with large pore-to-throat aspect ratios and under low flow rates, resulting in increased residual trapping during imbibition cycles (Krevor et al., 2015). In contrast, piston-like displacement can advance the wetting phase through stable fronts, thus suppressing snap-off and reducing H_2 trapping. These mechanisms are extremely sensitive to wettability, pore connectivity, and geometrical constraints (Hashemi et al., 2021; Kohanpur et al., 2022; Yu et al., 2024). Imbibition simulations in this study incorporated these key pore-scale mechanisms (*cf.* Fig. 2b-d), while the drainage was simulated following a piston-like displacement as described by invasion percolation algorithm (Valvatne and Blunt, 2004; Blunt, 2017).

While the quasi-static pore network modelling approach offers significant computational advantages, it inherently neglects dynamic effects such as viscous and inertial forces, which can become particularly important at high flow rates or in systems with complex wettability behaviour (Hosseinzadegan et al., 2023; Kohanpur et al., 2022; Blunt, 2017). Despite this limitation, it remains a reliable framework for simulating multiphase flow in porous media where capillary forces dominate fluid displacement and saturation evolution (Hashemi et al., 2021; Hashemi and Vuik, 2024), particularly in rocks such as those studied in this work. Under the assumptions of incompressible flow

Table 1
Fluid properties and wettability conditions considered for PNM of hydrogen-brine-basalt systems.

Condition	Contact angle (°)	Interfacial tension (mN/m)	Viscosity (Pa s)	Density (kg/m^3)
Strongly water-wet (Baseline)	$\theta = 19.2°$[a] $\theta_r = 0$–$35°$ $\theta_a = 12$–$28°$	65[a]	$\mu_{brine} = 5.97 \times 10^{-4}$ $\mu_{H_2} = 9.59 \times 10^{-6}$	$\rho_{brine} = 994.5$ $\rho_{H_2} = 7.2$
Weakly water-wet	$\theta = 62°$[a] $\theta_r = 0$–$35°$ $\theta_a = 50$–$67°$	55[a]	$\mu_{brine} = 5.97 \times 10^{-4}$ $\mu_{H_2} = 9.59 \times 10^{-6}$	$\rho_{brine} = 994.5$ $\rho_{H_2} = 7.2$

[a] Experimental data are sourced from Aftab et al. (2024).

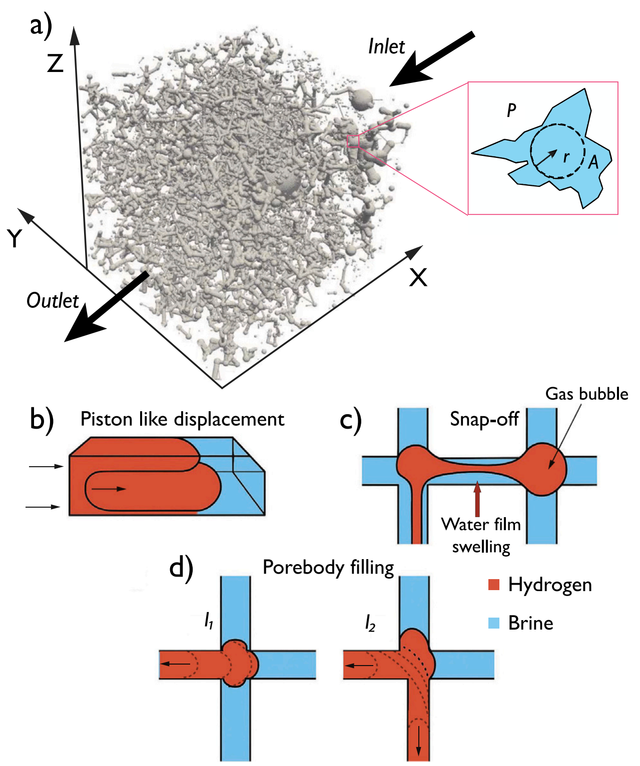

Fig. 2. Schematic for PNM setup and the respective drainage and imbibition fluid displacement mechanism. (a) REV geometry setup illustrating H_2 injection occurring at the inlet and displaced water exiting through the outlet. (b) non-wetting phase advances by piston displacement during drainage phase (c) snap-off due to wetting layer flow causing gas bubble formation (ganglion) (d) sequential pore body filling along with I_1 and I_2 events, during the imbibition process.

and a known distribution of fluid phases, local mass conservation is enforced at each network node as:

$$\sum_{j}^{N_i} Q_{ij} = 0 \tag{2}$$

where Q_{ij} is the volumetric flow rate between pore, i, and neighbouring pore body, j, and N_i is the coordination number, representing the number of pore throats connected to the pore body i. Further details on governing equations and the fluid-flow simulation procedure are provided in the Supplementary Text.

Wettability significantly influences capillary pressure at the pore level and thereby governs the spatial distribution of fluids (brine and H_2) within the pore network (Sun et al., 2020; Zhang et al., 2023). In this study, we consider a strongly water-wet system for the H_2-brine-basalt interactions, following experimental measurements by Aftab et al. (2024). The receding contact angle (θ_r), representing resident brine displacement during H_2 injection, and the advancing contact angle (θ_a), associated with brine re-entry during H_2 withdrawal, are listed in Table 1. Further analysis also includes less water-wet conditions, based on Aftab et al. (2024)'s observations of increased contact angles and reduced interfacial tension in the presence of sulfate-reducing bacteria. However, our model assumes that the change in contact angle and interfacial tension, representing the shift in wettability due to microbial activity, is established instantaneously and remains constant throughout subsequent drainage-imbibition cycles. As a result, potentially wettability-altering, time-dependent processes such as biofilm growth and decay, pore blocking, hydrogen consumption, and feedback between evolving saturation patterns and microbial dynamics are not fully captured, which may have implications for cyclic operations.

2.3. Benchmarking baseline model

Given the current lack of experimental data on H_2-brine multiphase flow behaviour in basaltic formations, our pore network and its model benchmarking was conducted by comparing predicted capillary pressure and relative permeability curves against the experimental results reported by Bertels et al. (2001) for fractured basalt rock. In their study, the intact basalt matrix exhibited an extremely low permeability on the order of 10^{-24}, indicating that fluid flow was primarily confined to fracture pathways and less likely through the rock pore matrix. Similar flow conditions were replicated by introducing fractures into the REV digital image as presented in Fig. 3, using a scaling procedure to adjust the fracture apertures reported for their 45 mm core to the dimensions (8.57 mm) of this study's sample. Further details on the geometric scaling approach are provided in the Supplementary Text.

In addition to the original unfractured and fractured pore networks (Fig. 3a and 3b), we generated two permeability-reduced variants using a binary morphological transformation to more closely replicate the intrinsically low permeability observed in the intact basalt core by Bertels et al. (2001). Specifically, a binary morphological closing operation was applied to the REV digital image, which suppressed smaller pore throats and isolated voids (Fig. 3c and 3d). This process reduced overall porosity and connected flow pathways, thereby decreasing the effective permeability of the network. The base REV digital images used for this transformation are illustrated in the Supplementary Text (*cf.* Fig. S4). Preliminary drainage simulations on the unfractured permeability-reduced network (Fig. 3c) failed to yield fluid displacement due to the absence of continuous, conductive pathways. The pore space in this case was too isolated and disconnected to support fluid invasion, rendering the network non-functional for multiphase flow analysis. Consequently, this network was excluded from subsequent simulations and discussion.

Drainage and imbibition simulations were carried out on the remaining pore networks using fluid properties matching those reported in Bertels et al. (2001). The resulting capillary pressure and relative permeability curves were then compared against their experimental measurements. It is important to note that their core-flooding experiment was conducted under higher flow rate conditions, corresponding to capillary numbers ranging from ~10^{-3} to ~10^{-7}. In these regimes, viscous forces play a substantial role in displacement behaviour, contrasting with the quasi-static assumptions of the pore network simulations. The key distinction is that the latter enforces mass conservation

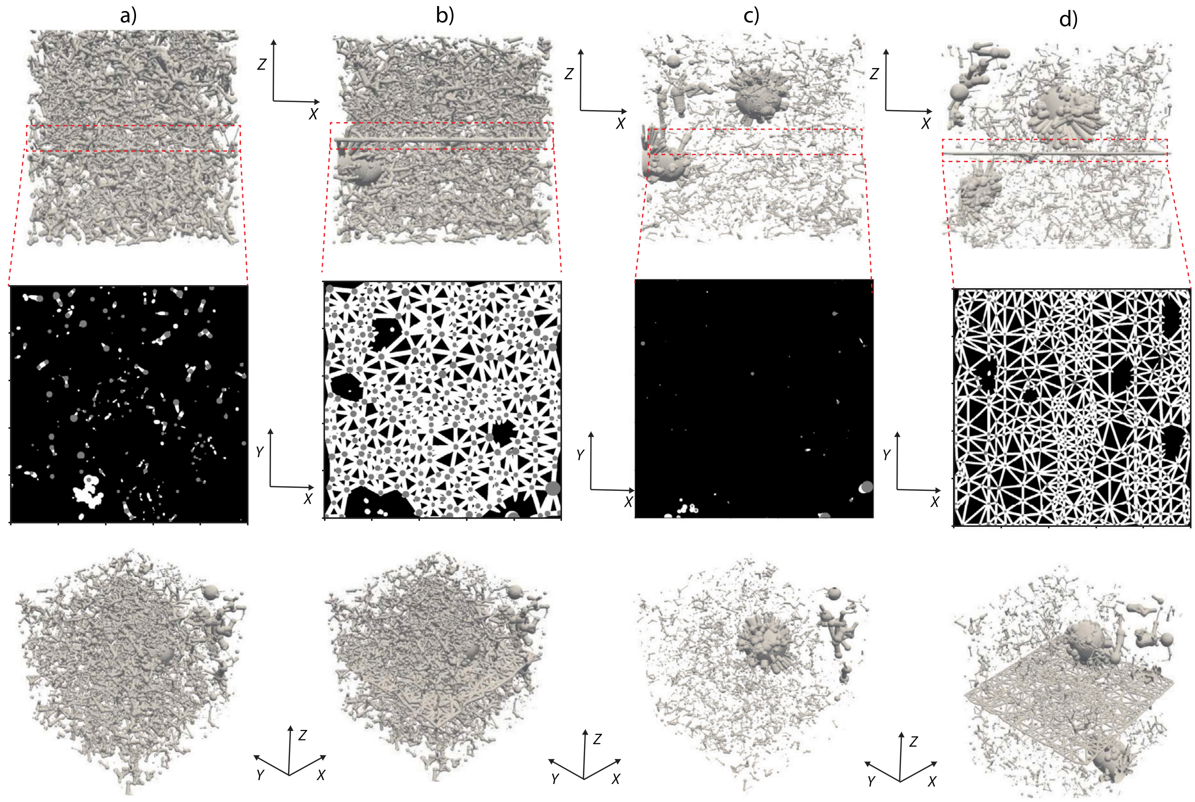

Fig. 3. Reconstructed basalt pore networks used for benchmarking and simulation. (a) Original network extracted from REV digital image (b) Fracture-induced network with artificial planar features introduced to investigate the influence of high-permeability conduits on flow behaviour. (c) Permeability-reduced network generated using binary morphological closing, which removes small pores and throats to replicate low-permeability characteristics (d) Fracture-induced permeability-reduced network combining fracture pathways with suppressed matrix connectivity to replicate the basaltic rock characteristics analysed by Bertels et al. (2001). The top panel shows XZ isometric views highlighting induced fracture locations; the middle panel shows the XY isometric view of vertical slice #500 to illustrate connectivity and morphological differences; and the bottom panel displays 3D framework of each pore network from a 1000^3 voxel domain (8.57 mm per side).

under equilibrium conditions, without accounting for flow rates or time-dependent saturation changes (Hashemi and Vuik, 2024). In this approach, the incoming capillary-driven flows into a pore body are summed up, neglecting viscous effects. In contrast, dynamic displacement experiments operate under non-equilibrium conditions, where the interplay between viscous and capillary forces becomes significant, which can potentially lead to viscous fingering phenomena (Li et al., 2025, 2017). These fundamental differences in flow regime were taken into account when interpreting the model-experiment comparison and evaluating the relevance of our pore-scale simulations to UHS in basaltic formations.

2.4. Initial-residual curve and recovery efficiency assessment

Quantifying capillary trapping during cyclic injection and withdrawal processes provides critical insights into residual hydrogen saturation and hysteresis evolution, which are essential for evaluating the efficiency and security of UHS (Lysyy et al., 2023; Khan et al., 2024). Trapping dynamics are commonly characterized using initial-residual (I–R) saturation relationships, which describe how much of the non-wetting phase (H_2) is retained in the pore space after imbibition, relative to the saturation it achieved during the preceding drainage cycle. A widely used empirical approach to characterize this behaviour is Land's trapping model (Land, 1968; Pini and Benson, 2017), which offer a practical framework for relating these saturation states. Land's model has been extensively applied in multiphase flow studies due to its relative simplicity and its effectiveness in capturing non-wetting phase trapping in porous media (Akbarabadi and Piri, 2013; Krevor et al., 2015; Pini and Benson, 2017). In the present work, Land's formulation was employed to interpret a series of simulated initial-residual saturation pairs derived from H_2-brine displacement within basaltic pore networks, as expressed in Eq. (3).

$$S_{nw,r} = \frac{S_{nw,i}}{(1 + C\, S_{nw,i})} \tag{3}$$

For each drainage-imbibition cycle, the initial H_2 saturation ($S_{nw,i}$) was defined as the maximum non-wetting phase saturation reached at the end of the primary drainage, while the corresponding residual saturation ($S_{nw,r}$) was determined at the end of the subsequent imbibition, and C is the empirical Land's trapping coefficient, ranging from 0 to infinity, reflecting the influence of rock type and wettability of the medium. After reaching a defined saturation point during drainage, the simulation proceeded through a full imbibition cycle to determine $S_{nw,r}$. This process was repeated across a range of drainage endpoints to build a set of I-R pairs, capturing the variation in trapping efficiency with increasing initial saturation.

In addition to these discrete I-R cycles, simulations were also conducted using cyclic H_2 injection and withdrawal schemes to evaluate how repeated cycling influences H_2 trapping and saturation hysteresis over time. These continuous cycles provide complementary insight into how capillary trapping evolves across successive operational cycles and are useful for informing long-term UHS performance assessments. Simulation-derived I-R data were further compared against literature-reported values for a range of H_2-brine systems to evaluate how the capillary trapping dynamics in basalt compares with those observed in other lithologies. This comparison highlights the influence of rock type on trapping behaviour and aids in situating basaltic systems within the broader context of UHS viability. To quantify storage performance, the corresponding UHS recovery efficiency for each drainage-imbibition

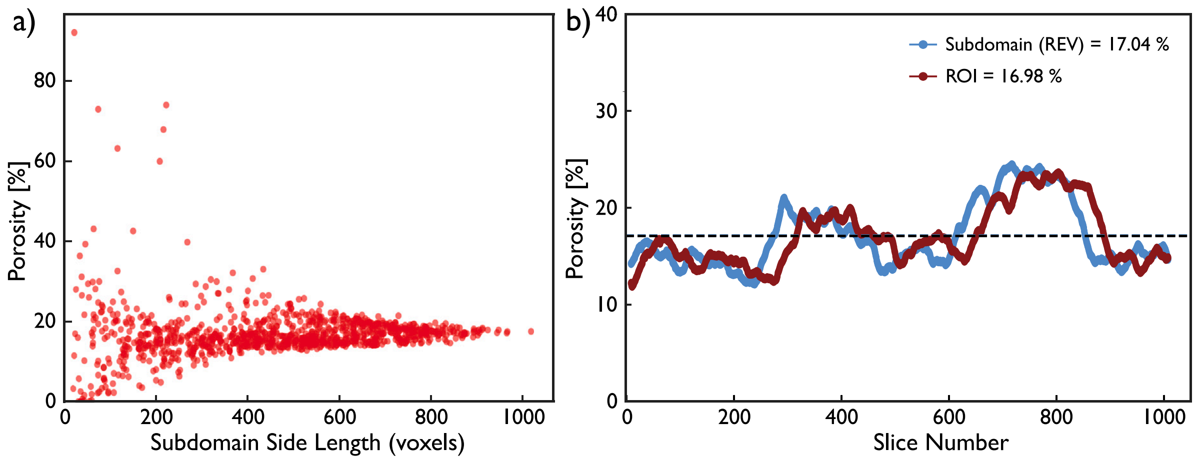

Fig. 4. Hawaiian basalt rock sample and subdomain porosity analysis. (a) Porosity variation across 1000 randomly sampled subdomains demonstrates convergence with increasing subdomain size; stabilizing around a side length of 1000 voxels. (b) slice-wise porosity comparison between the region of interest (ROI, dark red – ϕ_{avg} = 16.98%) and representative elementary volume (REV, blue – ϕ_{avg} = 17.04%). The dashed line indicates the average porosity 17.08% of the full core volume.

cycle was calculated using the following expression:

$$\text{Recovery Efficiency} = \left(\frac{S_{nw,max} - S_{nw,r}}{S_{nw,max}}\right) \times 100 \tag{4}$$

where $S_{nw,max}$ is the maximum H_2 saturation reached at the end of the drainage phase, and $S_{nw,r}$ is the residual saturation following imbibition. This metric offers a direct evaluation of how much H_2 can be recovered after each cycle, thereby serving as a critical performance indicator for assessing the feasibility of cyclic UHS operations in basaltic rocks.

3. Results and discussion

3.1. Geometric characterization of basalt pore networks

REV analysis was performed as described in Section 2 to determine a statistically representative subvolume suitable for quantitative characterization of the basalt pore network. Porosity variation was assessed across progressively larger voxel lengths as illustrated in Fig. 4. Due to the intrinsic heterogeneity of the basaltic sample, even with the selected ROI, significant variation in the mean porosity $\langle\phi\rangle$ was observed at small subdomain voxel length and a clear convergence was only achieved near the critical volume (V_{REV}) of $(1000, \text{voxels})^3$ (Fig. 4a). This subdomain volume was selected as the representative elementary volume for all subsequent pore-scale modelling and analyses. The porosity distribution within this REV is compared to that of the full ROI in Fig. 4b, confirming that the selected volume meets REV stability criteria while exhibiting an average porosity consistent with that of the ROI (16.98%) and full scanned volume (17.08%). This comparison further validates the adoption of this subvolume as the computational domain for subsequent image-based analyses.

Pore network extraction conducted using the maximal ball algorithm (Dong and Blunt, 2009; Raeini et al., 2019; Foroughi et al., 2025) decomposes the void space into discrete pores and throats, representing the wider void regions and the constrictions connecting them, respectively, with their resulting distribution presented in Figs. 5a and 5b. A distinct bimodal pattern is evident in the pore size distribution, driven primarily by the presence of large vug-like voids embedded within the basalt matrix (Figs. 1a and 3a; see Table S2 for detailed statistics). Visual and quantitative assessment suggests classification of pores into two size populations: small pores with diameters less than 85 µm, and large pores beyond this threshold, with relative frequency peaks near 55 µm and 125 µm, respectively. Notably, the pore sizes resolved are substantially larger than the voxel resolution (8.5 µm/voxel), ensuring that these observations reflect true pore-scale features rather than imaging artifacts. The small-pore population (≤85 µm) exhibits a high and relatively uniform frequency, indicative of a densely connected pore network likely essential for baseline permeability and connectivity. In contrast, the large-pore population comprises a mixture of intermediate and prominent vug-like features that can act as preferential pathways for high-permeability flow. The corresponding throat size distribution (Fig. 5b) is unimodal and more uniform in range, reflecting consistent constriction geometry across the network.

While the precise pore structure in basalts can vary depending on emplacement conditions, cooling rate, and post-eruptive alteration, bimodal pore size distributions are not uncommon (Baker et al., 2012). Many basaltic rocks, particularly those with vesicular textures or altered flow top zones, exhibit both fine-scale pores and larger voids or fractures (Shin et al., 2005). Hence, although the distribution observed here reflects the unique characteristics of this sample, it is broadly consistent with textural heterogeneity reported in other studies (Shin et al., 2005; Awolayo et al., 2022; Stavropoulou et al., 2024). Together, these patterns highlight the multiscale heterogeneity of the basaltic pore space, where fine-grained pore clusters and larger vugs coexist and are uniformly interconnected. Such a configuration has important implications for flow and transport behaviour, particularly under multiphase transport conditions where capillary and diffusive effects are strongly scale-dependent.

Comparisons with the Hawaiian picrite basalt studied by Stanfield et al. (2024) reveals notable petrophysical differences. They reported a higher permeability (42.9 D) and porosity (25.5%) than those observed in this work (452.83 mD and 17.04%, respectively) (Table S1). The nearly two-orders-of-magnitude difference in permeability likely stems from a higher abundance of relatively smaller pores and narrower throats in this study's sample, which limit fluid flow and reduce bulk transport capacity. The effective porosity derived from the extracted network is 15.1%, suggesting the presence of isolated or poorly connected pores. Such disconnection may reflect intrinsic pore-scale isolation or be an artifact of image resolution, which can obscure narrow connections. This limitation is particularly relevant given the voxel size of 8.57 µm, which reflects the classic trade-off between resolution and field of view. Imaging a centimetre-scale volume large enough to satisfy the REV requirement necessitates a coarser voxel size, potentially missing sub-voxel features (Sietins et al., 2022; Carrillo et al., 2022). In contrast, pushing to sub-micron resolution would restrict the field of view to only a few millimetres, rendering the scan statistically unrepresentative. These unresolved micropores at coarser voxel sizes can influence multiphase flow behaviour in several ways. If well connected, they may act as percolation pathways that keep the

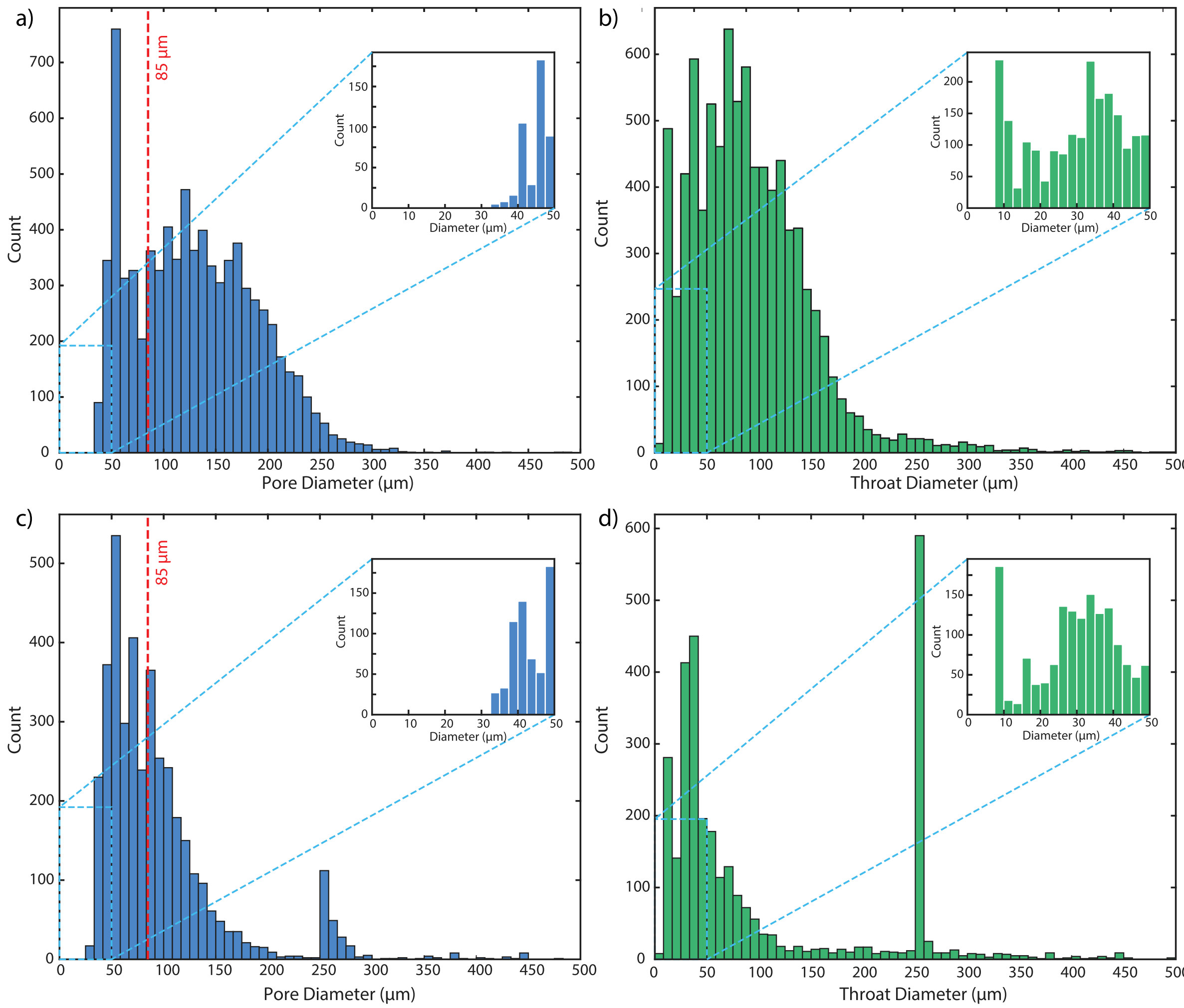

Fig. 5. (a) Pore size distribution (PSD) and (b) throat size distribution of the extracted pore network, showing a bimodal pattern characteristic of structural heterogeneity. (c) PSD of the basaltic network following fracture introduction, highlighting notable pore rearrangement and a more pronounced bimodal distribution. (d) Throat size distribution post-fracture, illustrating rearrangement effects and the emergence of new peaks associated with the fracture-induced pathways.

network conductive even at low water saturations, allowing brine to drain from large vugs. Conversely, if poorly connected, they function as dead ends that trap brine, reduce gas mobility, and diminish displacement efficiency. Quantifying this unresolved fraction represents a critical uncertainty, especially in bimodal or micro-fractured systems like basalt.

Further geometric characterization of the extracted network was carried out on the REV subvolume shown in Fig. 3a, capturing key attributes of the heterogeneous basaltic pore structure and their pore-scale variability in shape factor ($G = R^2/4A$), pore-throat aspect ratio, and coordination number (*cf* Fig. S3 and Table S2). The distribution of these features as functions of pore diameter in Fig. 6 highlights the geometric and topographic variability that may govern hydrogen trapping and displacement behaviour within basalt. The pore shape factor remains low and relatively constant for pores smaller than 100 μm, indicating a predominance of angular and irregular geometries in small pores, typical of fine-grained basaltic textures (Singh et al., 2022). Beyond this threshold, the shape factor gradually declines, suggesting increasingly non-circular cross-sections in larger pores (Fig. 6a). Throat shape factors also decrease with increasing pore diameter for larger pores, implying that throats connected to larger pores are more angular and geometrically complex. The aspect ratio, defined as the pore radius relative to the connecting throat radius, exhibits a non-monotonic trend across pore size classes (Fig. 6c). It is relatively high in small pores (≤85 μm), decreases to a minimum in intermediate pores (86 - 200 μm), and then increases again in the larger pore class (≥500 μm). This U-shaped distribution suggests distinct pore-throat scaling behaviours at across pore size ranges. The initially high aspect ratio may reflect irregular, angular small pores with disproportionately small throats, while the increase in the larger pores likely arises from vug-like features connected by narrow throats.

Coordination number, representing the number of throats connected to each pore, increases systematically with pore diameter (Fig. 6d). Small pores (≤85 μm) exhibit limited connectivity, with coordination numbers below 1. Connectivity then gradually increases, with the largest pores (≥500 μm) standing out due to a coordination number exceeding 30. This sharp contrast indicates that large pores act as highly connected nodes within the pore network, potentially dominating bulk transport. The disparity in connectivity may lead to significant flow heterogeneity and localized trapping during fluid displacement processes. Collectively, these geometric trends reveal a complex, multi-scale structure characterized by embedded vugs, narrow throats, and highly connected large pores that can strongly influence hydrogen displacement efficiency, capillary trapping, and snap-off behaviour. These implications are further explored through two-phase flow simulations under both fractured and unfractured conditions in the following sections.

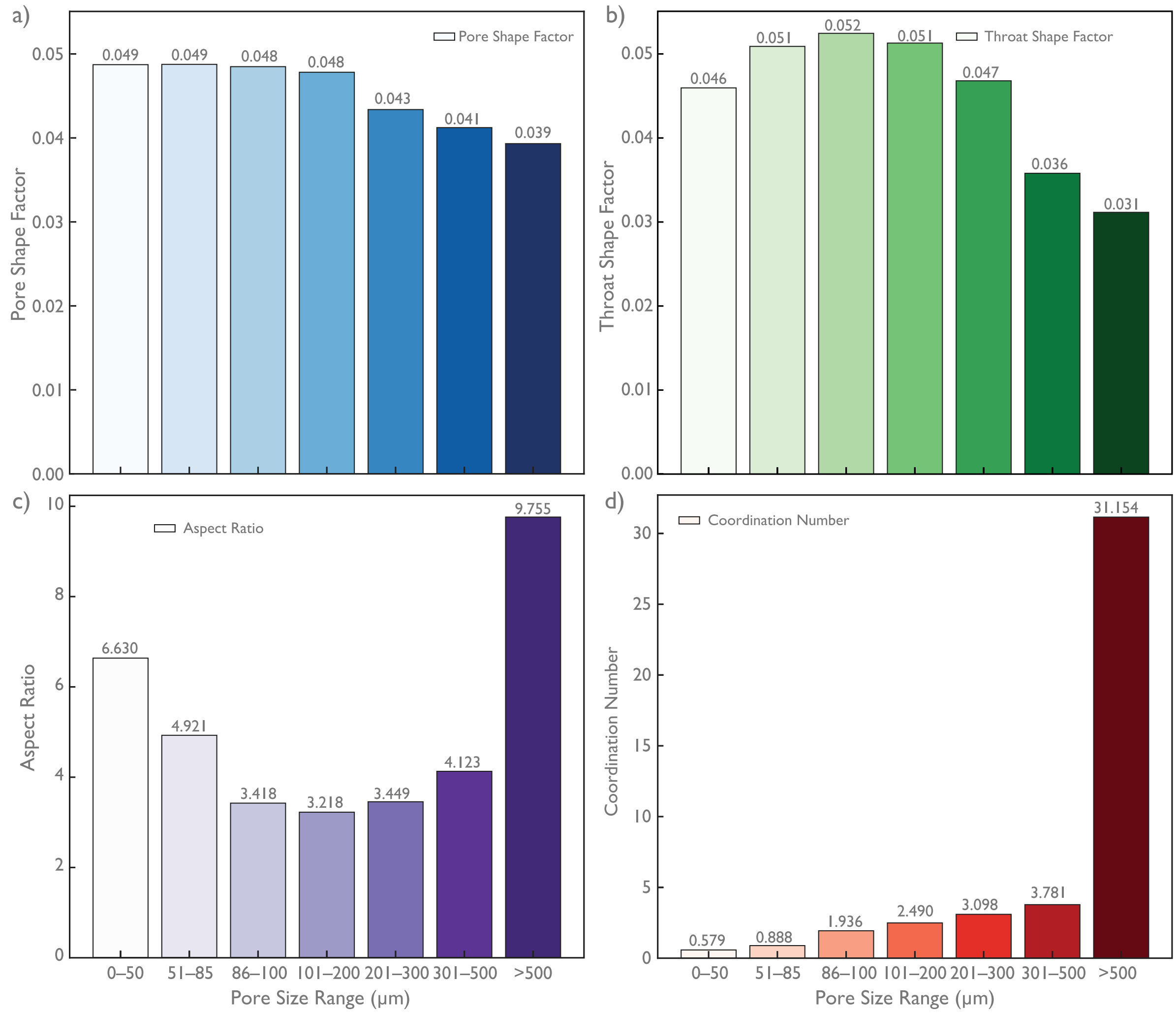

Fig. 6. Distribution of pore network geometric features as a function of pore diameter, including: (a) pore shape factor showing relatively consistent low values for pores smaller than 100 µm, (b) throat shape factor decreasing with increasing pore diameter for larger than 100 µm, (c) aspect ratio showing minimal variation between small and intermediate pore sizes but rises substantially for pores ≥ 500 µm. (d) coordination number increasing with pore diameter range.

3.2. Benchmarking PNM with fractured and unfractured basalt samples

Following pore network extraction, PNM simulations were benchmarked against the experimental data from Bertels et al. (2001) in Fig. 7 and Fig. S5 to evaluate the model's predictive capabilities. A synthetic fracture was introduced into the basalt pore network, resulting in a noticeable increase in pore connectivity from 2.13 in the unfractured network to 2.3 in the fractured network, representing the upper bound of the scaled aperture range (Table S4). The representative mean fracture aperture was estimated to be ∼10 ± 7.5 voxels (Table S3), however, simulations were conducted across a range of 10 to 35 voxels to evaluate sensitivity to aperture size. This modification also altered the pore and throat size distributions, as shown in Figs. 5c and 5d, with a higher frequency of larger pore throats reflecting enhanced overall connectivity due to the synthetic fracture. Preliminary flow simulations through the fractured network confirmed a significant increase in permeability, driven by the formation of highly conductive channels and improved pore connectivity (Table S4). The resulting relative permeability curves (Fig. S5) showed a pronounced sharp decline in water relative permeability (k_{rw}) and a steep increase in gas relative permeability (k_{rg}) at low gas saturations. These trends deviate substantially from the experimental data and reflect characteristic behaviour of fracture-dominated systems, where gas preferentially bypasses the brine-saturated matrix through well-connected, high-permeability pathways.

This mismatch in the transport behaviour prompted a morphological adjustment of the fractured pore network, which reduced pore connectivity and permeability (*cf.* Table S4). The revised (permeability-reduced) fractured network was evaluated against both the unfractured and experimental characteristic curves in Fig. 7. The simulation employed a fracture aperture of 35 voxels, with sensitivity analysis spanning apertures from 0 (unfractured) to 35 voxels to define confidence bounds. The resulting envelope captures the influence of fracture aperture on both relative permeability and capillary pressure curves, providing interpretive bounds on the experimentally observed fracture-controlled behaviour (Bertels et al., 2001). Fig. 7a shows that the simulated relative permeability curves from the fractured network reproduce the general trends observed in the experimental data of Bertels et al. (2001), particularly for the wetting phase (k_{rw}). However, k_{rg} is overestimated, with significant gas flow occurring even at high water saturations, indicative of preferential flow through the synthetic fracture. In contrast, the unfractured network yielded lower k_{rg} that more closely matched the experimental trend, reflecting the limited gas mobility characteristic of an intact, low-permeability matrix.

Capillary pressure comparisons (Fig. 7b) further support the model's validity. Simulated pressure-saturation curves for the fractured network align well with experimental observations, particularly at high water saturations. Some deviations at intermediate saturations suggest that the capillary-dominated PNM approach is unable to fully

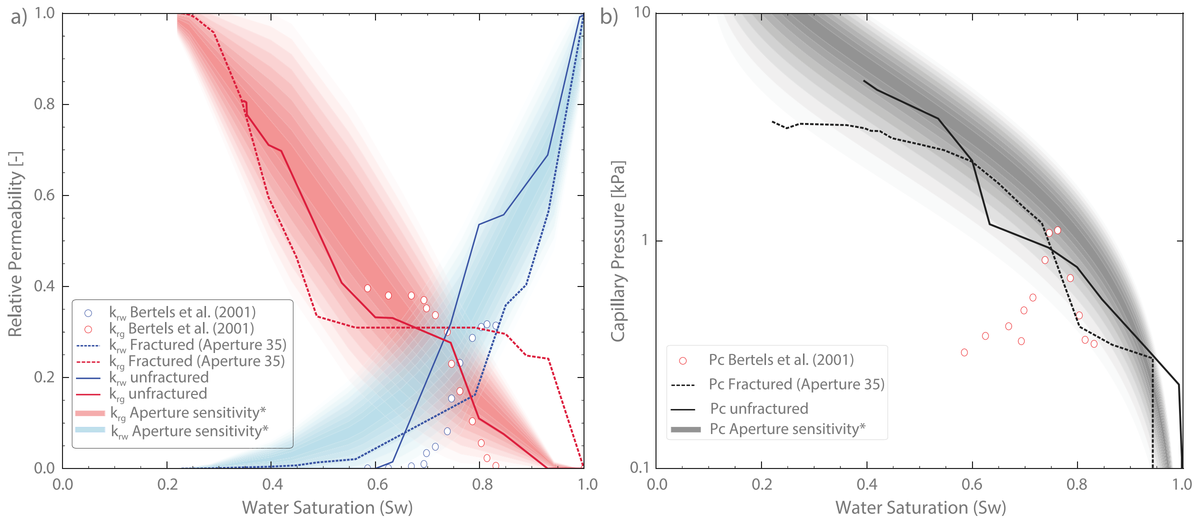

Fig. 7. Comparison of simulated (a) capillary pressure curves and (b) relative permeability of unfractured and fractured basalt simulation with experimental data from Bertels et al. (2001). The shaded region, labelled as *Aperture sensitivity**, represents simulation sensitivity bounds at a 95% confidence interval for aperture sizes ranging from 10 to 35 voxels, highlighting the effect of aperture size on characteristic flow curves.

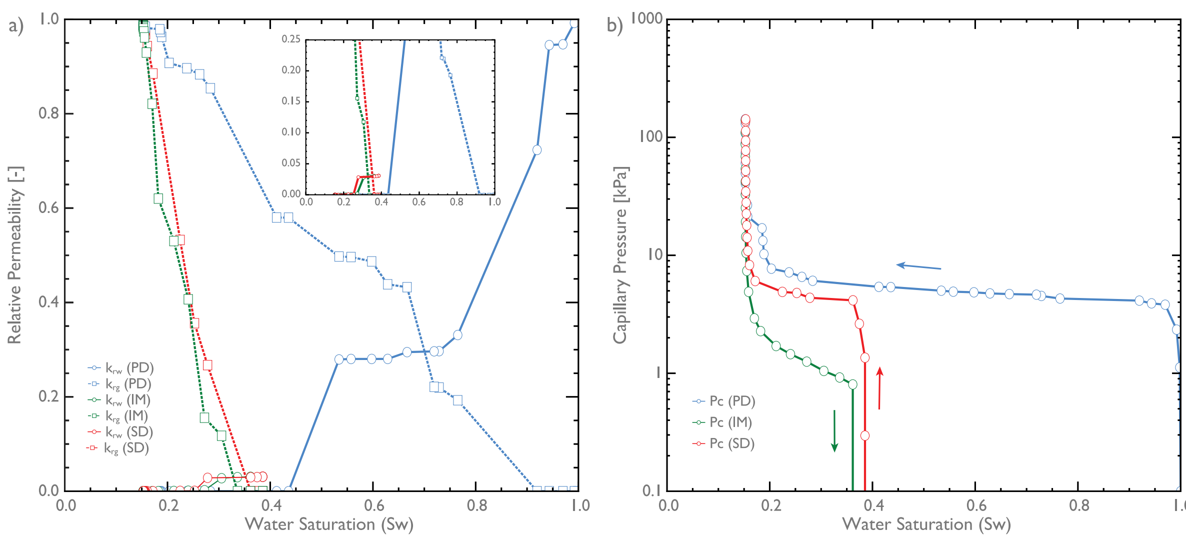

Fig. 8. Hysteresis in (a) relative permeability and (b) capillary pressure curves for H_2–brine–basalt pore system. In the drainage (PD) process, the rock is initially saturated with brine, and H_2 is injected as the invading fluid. Endpoint maximum H_2 saturation ($S_{g,max}$) is reached when the capillary pressure peaks at 142 kPa, and brine permeability approaches zero, corresponding to the residual brine saturation (S_{wr}). In the subsequent imbibition (IM), brine re-enters the pore space, displacing H_2 as water saturation increases, until a maximum capillary pressure of 10 kPa is reached, at which point further brine invasion immobilizes H_2 phase transport. The capillary pressure curve during IM sharply declines over a narrow saturation range, diverging from drainage path due to capillary trapping, while secondary drainage (SD) reveals altered flow pathways shaped by prior saturation history.

reproduce the viscous effects and surface roughness inherent in the physical experiments (*cf.* Section 2.3). Nevertheless, the observed trend of decreasing capillary entry pressure with increasing fracture aperture is accurately captured, demonstrating the model's sensitivity to geometrical variations. The shaded confidence envelope quantifies this sensitivity, enhancing the robustness of the comparison. While neither the fractured nor unfractured network models fully replicate the experimental trends on their own, their combined sensitivity envelope closely brackets the observed behaviours. This suggests that the key lies in the distribution of variable fracture apertures, an aspect prominently reported by Bertels et al. (2001) but not fully captured in our current digital reconstructions. The results imply that if this geometric complexity were more accurately represented, particularly the distribution of fracture apertures variability, full replication of the experimental trends may be achievable. Collectively, these findings validate the PNM framework for simulating multiphase flow in basalt and underscore the importance of accurately characterizing fracture geometry. Building on this framework, subsequent analyses applied the unfractured basalt network to investigate cyclic multiphase flow processes, including primary drainage, imbibition, and secondary drainage, with a focus on saturation evolution and hysteretic flow behaviour.

3.3. Impact of hysteresis on relative permeability and capillary pressure

During primary drainage (PD), the relative permeability and capillary pressure trends reflect the transition from a brine-saturated system to one increasingly dominated by H_2 gas flow (Fig. 8). At the onset of PD, the brine phase exhibits maximum connectivity and mobility (k_{rw} = 1 at S_w = 1), while the gas phase remains disconnected. As H_2 begins to invade, it preferentially enters the largest pores under relatively low capillary pressure (Fig. 8b), rapidly establishing connected pathways and confining brine to smaller, less permeable pores. This invasion is captured in Fig. 9, which illustrates the dynamic evolution of brine displacement by H_2 through the basalt pore network.

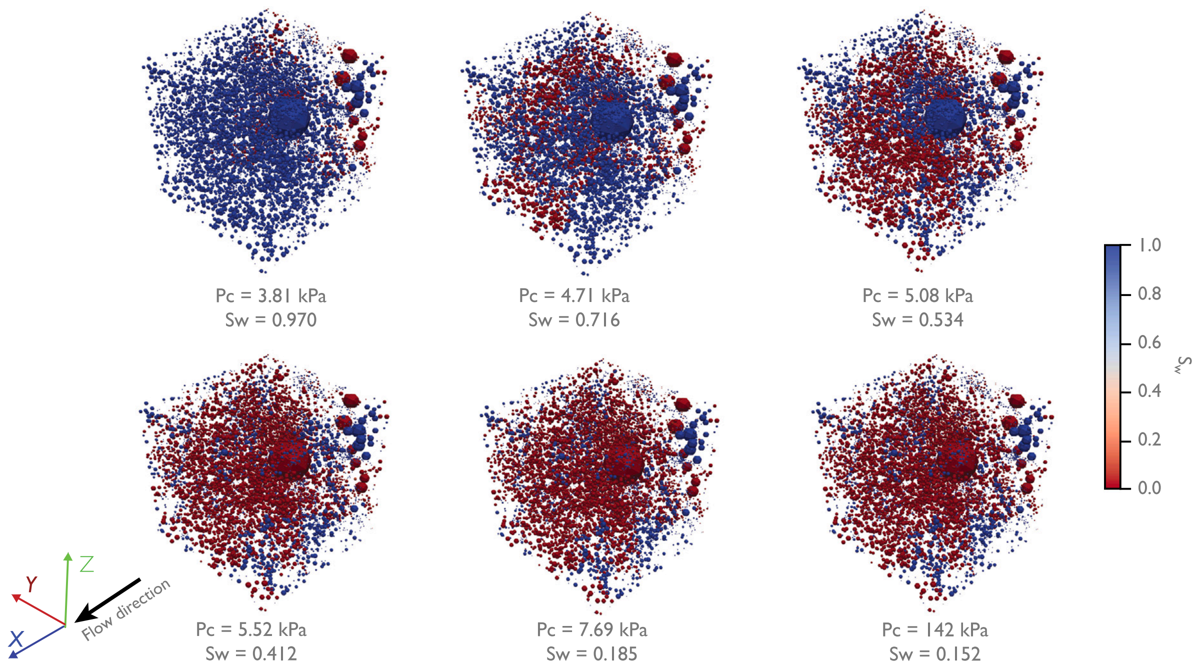

Fig. 9. Evolution of spatial distributions of phase saturations in the basalt pore network during primary drainage (PD). The wetting-phase fluid (brine, blue) is progressively displaced by the non-wetting-phase fluid (H_2, red) as capillary pressure increases, resulting in decreased wetting-phase saturation and increasing H_2 gas connectivity.

The initial gas percolation occurs at a low capillary pressure of 3.81 kPa, when the wetting phase still occupies nearly the entire pore volume ($S_w \approx 0.97$). As capillary pressure increases, H_2 advances into progressively small pores, enhancing its network connectivity and mobility (Fig. 9). This process coincides with a rapid increase in k_{rg}, even at low gas saturations, as H_2 preferentially occupies larger, better-connected pores, facilitating the formation of continuous gas channels and substantial amount of gas connected within the network (Fig. 9). By contrast, brine becomes increasingly restricted to isolated or poorly connected pores, leading to a sharp decline in k_{rw} (Fig. 8a). A crossover point, where $k_{rg} = k_{rw}$, occurs at $S_w \approx 0.71$, consistent with a strongly water-wet (hydrophilic) system (Lysyy et al., 2022). As drainage proceeds and the capillary pressure approaches 142 kPa, brine saturation continues to decrease, and gas flow becomes dominant. Gas relative permeability reaches a peak value near-one, while water relative permeability drops to near-zero due to disconnection and confinement in the smallest pores. At the end of PD, the brine phase is reduced to a residual saturation ($S_w \approx 0.15$), trapped within the rock matrix and no longer contributing to flow.

During imbibition (IM), the pore network filled with H_2 begins to resaturate with brine. As the wetting phase re-enters the pore space, it preferentially invades smaller pores first, driven by higher capillary pressures (Figs. 8b and 10). Relative permeability trends (Fig. 8a) reveal a gradual increase in brine mobility as its saturation increases. At $S_w = 0.390$, k_{rw} rises to ~0.031, while the H_2 phase becomes disconnected, leading to a sharp reduction in H_2 mobility. This transition corresponds with elevated capillary pressures (>10 kPa) observed at low water saturations (Fig. 8b), indicative of the strong capillary forces required to displace the non-wetting phase from finer pore structures. As capillary pressure decreases during imbibition, ranging from 142 kPa (drainage endpoint) to −10 kPa (imbibition regime), water saturation increases from 0.15 to 0.39, reflecting the progressive invasion of brine into the pore network. The saturation endpoint of residual gas, $S_{gr} \approx 0.61$ indicates the trapped H_2 within the pore structure, a consequence of capillary trapping and snap-off events that inhibit reconnection in pore throats.

The behaviour of k_{rg} and k_{rw} across the drainage-imbibition cycle further highlights the hysteretic nature of the pore system. Specifically, k_{rg} values are highest during primary drainage, lower during secondary drainage, and lowest during imbibition, consistent with gas-water relative permeability measurements in strongly water-wet systems (Lysyy et al., 2022). This trend is attributed to residual gas trapping during imbibition, which limits the re-establishment of continuous H_2 gas pathways. The elevated residual gas saturation highlights the extent to which the pore space remains disconnected from the primary flow paths accessible during drainage, highlighting the controlling role of pore-scale heterogeneity and capillary entrapment on multiphase transport in basaltic formations.

3.4. Phase occupancy dynamics and trapping behaviour

Building on the preceding analysis of hysteresis in saturation trends, we examine the distribution of fluid phases across pore sizes to better understand the mechanisms contributing to elevated residual trapping. PNM predictions of pore-by-pore fluid occupancy, though sensitive to calibration, have shown reproducibility within 10%–20% uncertainty (Raeini et al., 2019; Foroughi et al., 2021). A conservative 10% uncertainty margin is therefore adopted in Fig. 11 to aid interpretability, which illustrates the pore-scale occupancy patterns of brine and hydrogen (H_2) during both drainage and imbibition. During drainage, H_2 preferentially invades larger pores (>85 μm) due to their lower capillary entry pressures (Figs. 8b and 9), displacing brine and establishing high non-wetting phase saturation in these regions. Meanwhile, brine is retained predominantly in smaller pores (≤85 μm), consistent with entry pressure constraints and brine's wetting affinity (Fig. 11a). These trends align with observations by Thaysen et al. (2023), who reported that H_2 typically occupies pore centers, while brine resides in corners, throats, and as thin films along grain surfaces. A different pattern emerges during imbibition (Fig. 11b), where brine reoccupies much of the smaller pore volume, increasing its saturation within the 0–85 μm range. However, H_2 becomes increasingly trapped as clusters in larger pores (≥86 μm), particularly those in the 100–300 μm range, where hydrogen saturation remains dominant (Fig. 11b). This reflects strong capillary entrapment governed by the geometric and connectivity features of the basalt pore network.

This partitioning of fluid phases and size-selective trapping is intricately linked to the underlying pore geometry, as further clarified

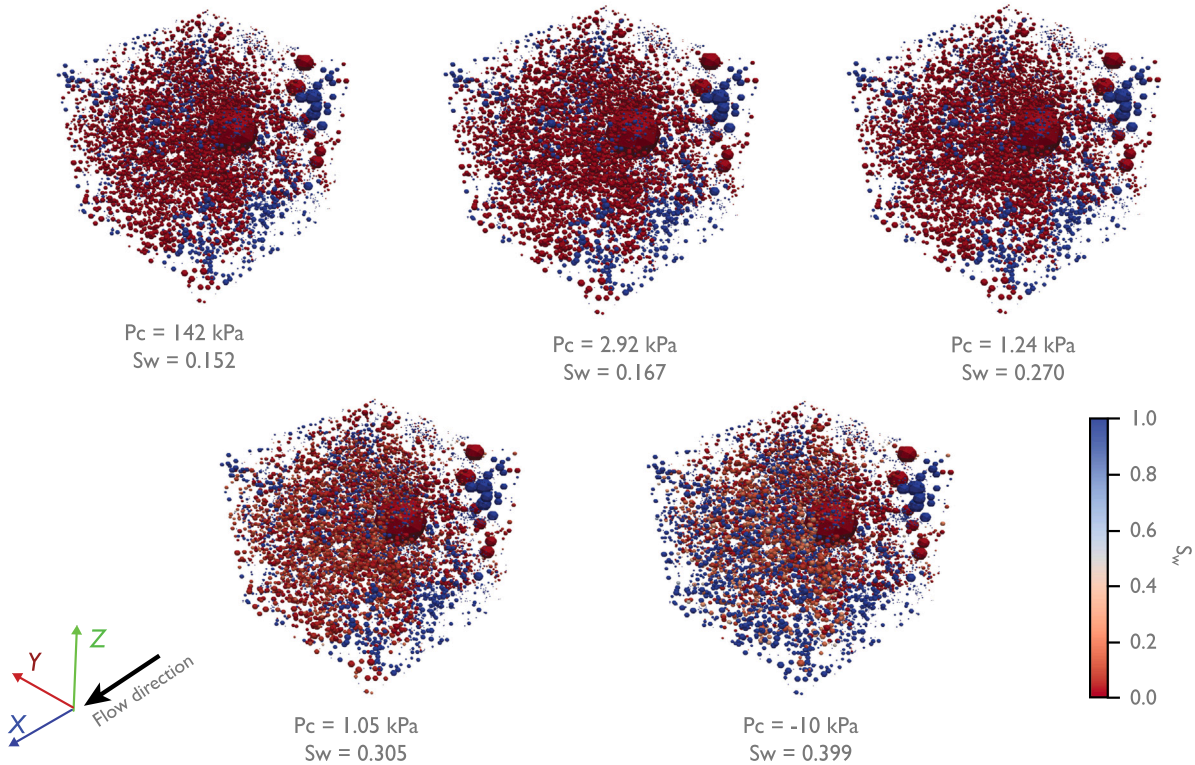

Fig. 10. Evolution of spatial distributions of phase saturations in the basalt pore network during imbibition. The non-wetting-phase fluid (H_2, red) is progressively displaced by the invading wetting-phase fluid (brine, blue) as capillary pressure decreases, resulting in increased brine saturation and gradual disconnection of H_2 pathways due to capillary trapping.

by the volume-weighted pore occupancy analyses (Figs. 11c and 11d). During drainage, H_2 dominates the 51–200 μm range, occupying up to 80%–85% of the pore volume, while brine is confined primarily to smaller pores, comprising 52% of the 0–50 μm range (Fig. 11c). These intermediate pores offer moderate throat sizes and connectivity, making them accessible to the non-wetting phase while preserving sufficient capillarity for efficient brine displacement. Smaller pores (0–50 μm), which are abundant and angular (as indicated by high shape factors, Figs. 6a and 6b), exhibit the lowest occupancy by H_2 during drainage. These pores have disproportionately small throats, low coordination numbers, and high shape factors conducive to the formation of stable corner wetting films. These attributes promote wetting film stability and impose capillary entry barriers, limiting non-wetting phase entry even under elevated capillary pressures and thereby promoting brine retention. Although such pores are more readily reoccupied by brine due to their small size and high entry pressures, their limited interconnectivity impedes the efficient displacement of the non-wetting phase (Fig. 11d). The largest pores (≥500 μm) exhibit the highest hydrogen occupancy (100% saturation), likely due to their frequent connection through narrow throats.

Following imbibition, brine readily displaces H_2 from much of the small to intermediate pore spaces, increasing saturation to 71% in the 0–50 μm range (Fig. 11d). Nonetheless, H_2 remains trapped in the larger size classes, retaining 70% and 100% saturation in the 301–500 μm and ≥500 μm pores, respectively. These quantitative estimates confirm that trapping is most pronounced in specific pore size windows and that full displacement of H_2 is not achieved despite brine re-entry. The observed efficiency in reducing H_2 occupancy from intermediate-sized pores is attributable to their relatively balanced aspect ratios and moderate connectivity, which allow for effective rewetting and hydrogen displacement. In contrast, the largest pores (≥500 μm) display markedly different geometrical features that predispose them to capillary snap-off. Their coordination numbers exceed 30 (Fig. 6d), indicating a high degree of interconnectivity, while their aspect ratios reach values up to 9.76 (Fig. 6c). These high aspect ratios highlight a pronounced disparity between pore bodies and narrower connecting throats, creating favourable conditions for snap-off, particularly when throat constrictions impede continuous phase displacement.

This observation aligns with the findings of Peng et al. (2025), who reported that snap-off is especially prevalent where the pore-to-throat radius ratio exceeds 2. Singh et al. (2022) further demonstrated that aspect ratio exerts dominant control over snap-off propensity, with a critical transition range of 1.4–2.15 identified in Ketton limestone. The larger aspect ratios observed in basalt pores exceed this threshold, suggesting a higher likelihood of non-wetting phase entrapment. In addition, their mean shape factors (0.039 for pores and 0.031 for throats, Fig. 6a–b) indicate predominantly angular and irregular geometries. Such low shape factors are conducive to the development and stabilization of corner wetting films, especially during imbibition. Narrow corner angles and large interfacial areas within pore throats support the residence and swelling of wetting layers under reduced capillary pressure or increased brine pressure. These layers may expand and abruptly snap off non-wetting phase clusters, resulting in significant gas trapping, and such snap-off events can significantly influence residual gas saturation (Krevor et al., 2015). Notably, the throat shape factors approach the critical threshold of 0.026 reported by Singh et al. (2022), below which snap-off becomes more favourable due to enhanced film swelling and throat angularity.

The combined influence of low shape factors, high aspect ratios, and varied coordination numbers indicates that larger basalt pores are particularly susceptible to capillary snap-off and the formation of disconnected hydrogen clusters. These geometric constraints, together with the dynamics of imbibition, drive the observed size-selective trapping behaviour and the retention of residual gas saturations within specific pore size intervals. Small pores limit flow due to restricted connectivity and high entry pressures, while larger pores act as snap-off-prone traps owing to their geometric disproportion and wetting film

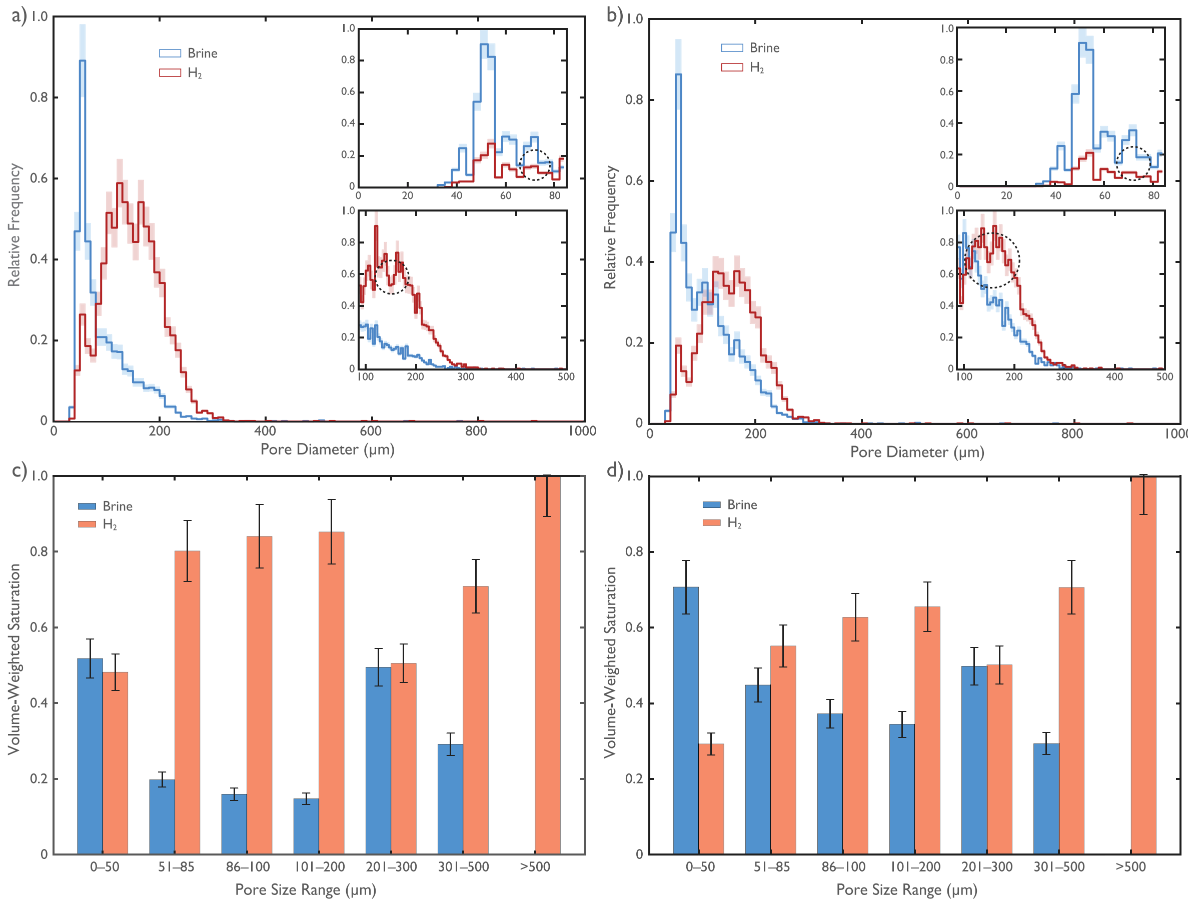

Fig. 11. (a) Phase occupancy distribution at the end of (a) primary drainage (PD) and (b) secondary imbibition (SI). Corresponding volumetric pore occupancy for brine (wetting phase) and H_2 (non-wetting phase) at the end of (c) PD and (d) IM stages. Expanded views of the phase occupancy in *a* and *b* are compared for small pores (0–85 μm) and large pores (>85 μm), illustrating preferential H_2 filling in larger pores during drainage and brine re-invasion during imbibition. The full distribution is computed using a 10 μm bin size; expanded views use bin sizes of ~2.93 μm (small pores) and ~4.61–5.93 μm (large pores). Shaded bands and error bars denote ±10% uncertainty.

dynamics. This dual control highlights the critical interplay between pore structure and phase displacement efficiency under cyclic flow conditions, underscoring the importance of geometric metrics in predicting entrapment behaviour in basalt networks. Understanding this size-selective trapping mechanism is critical for optimizing recovery strategies and informing predictive models of hydrogen migration in basalt reservoirs.

3.5. Hysteresis-driven saturation evaluation

Understanding the relationship between initial and residual hydrogen saturation is essential for evaluating H_2 trapping efficiency during UHS operation. Initial–residual (I-R) saturation curves were generated for varying initial saturations, as shown in Fig. 12. As the initial hydrogen saturation increases, the residual saturation correspondingly increases, indicating that greater initial occupancy leads to more extensive phase trapping. This trend reflects enhanced H_2 retention during imbibition, with the residually trapped H_2 strongly influenced by the pre-imbibition saturation state. The Land's trapping coefficient (C), calculated using eq. (3), quantifies this relationship, with lower values of C indicating higher residual trapping. At higher initial saturations, a larger portion of the pore network is occupied by the non-wetting gas phase, increasing the likelihood of capillary trapping as the brine re-enters during imbibition, particularly due to enhanced snap-off events in narrow pore throats, angular pore geometries with swelling wetting layers, increased gas phase discontinuity, and the formation of isolated ganglia within poorly connected pore regions (*cf.* Section 3.4).

Although I-R saturation data for basalt lithologies is limited, our results were compared to the findings of Khan et al. (2024), who reported C values ranging from 0.2 to 2.7 across different sedimentary rock types. The C values obtained in this study (0.5 to 1.4) fall within this range and notably align with values observed for sandstones (Higgs et al., 2024; Boon et al., 2024b; Thaysen et al., 2023; Lysyy et al., 2023), but are generally lower than those reported for carbonate rocks (1.32 to 3.63), suggesting a comparatively greater H_2 trapping capacity in basalt. The scattered deviations in comparison with H_2 storage in sandstone reservoirs in Fig. 12 is likely due to differences in pore connectivity (Figure S6), wettability, and geometrical attributes. Additionally, Khan et al. (2024) reveals that increasing the receding contact angle, which is typical at deeper conditions, shifts C range higher (from 0.2–2.7 to 0.3–3.2), indicating reduced trapping efficiency in deeper, less water-wet conditions. This observation is significant considering that basalt formations may be accessible at shallower depths than conventional sedimentary UHS targets. At such shallower depths, where overburden pressure and temperature are lower, the stronger capillary forces associated with basalt's pore structure can contribute to more H_2 trapping. This comparative analysis across lithologies, depth regimes, and pore-scale attributes demonstrates that basalt, despite its limited prior investigation, exhibits strong capillary trapping behaviour that could support secure and efficient hydrogen recovery, particularly in shallow subsurface settings where capillary forces dominate, provided that appropriate engineering strategies are employed to manage and optimize its displacement dynamics.

We also evaluated how cyclic injection and withdrawal impact H_2 trapping over successive drainage-imbibition cycles. These cycles revealed oscillations in residual saturation, reflecting the dynamic redistribution of fluid phases and evolving pore occupancy. Fig. 13

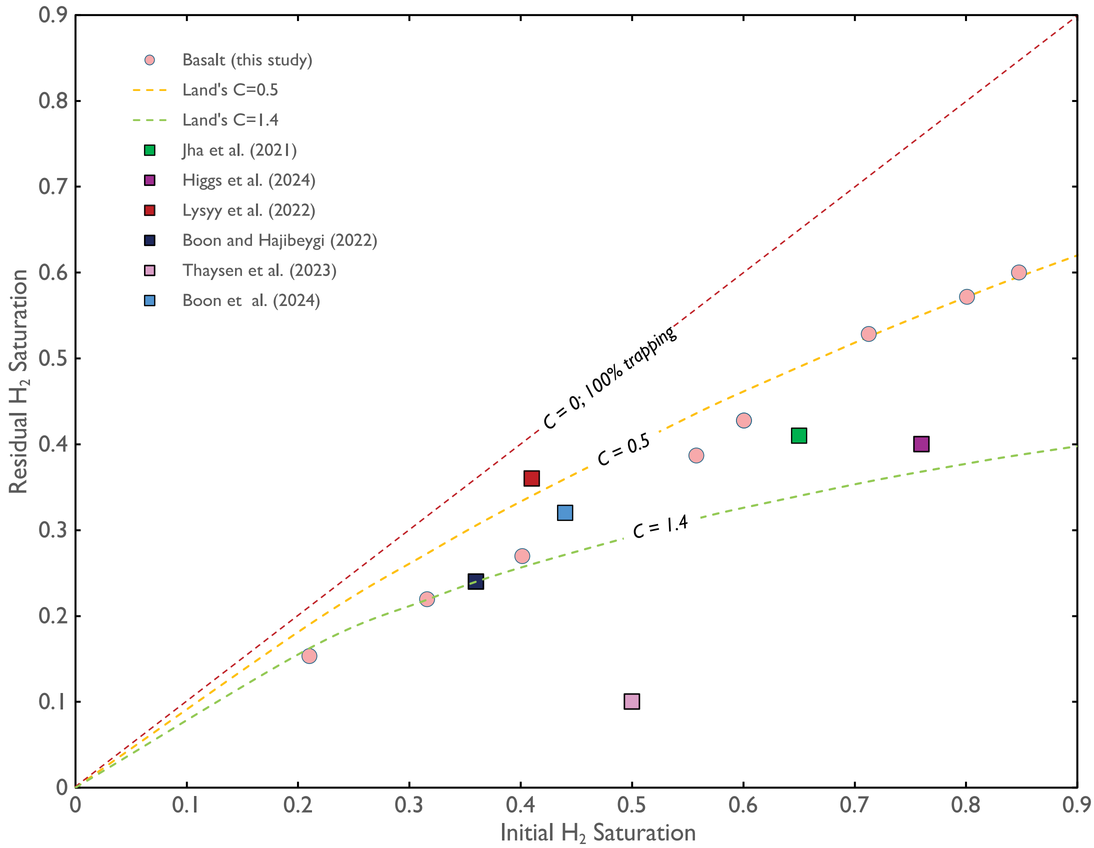

Fig. 12. Trapped H_2 saturation as a function of initial H_2 saturation, comparing results from this study (basalt) with published experiments on sandstone rocks, including Berea (Jha et al., 2021; Boon et al., 2024b), Bentheimer (Higgs et al., 2024), Clashach (Thaysen et al., 2023), and Fontainebleau (Lysyy et al., 2022). Dashed curves show Land's empirical trapping model for various C values. Basalt data align closely with the $C = 0.5$ curve, indicating intermediate trapping behaviour. In contrast, Fontainebleau and Clashach sandstones follow higher C values ($C = 1.4$), reflecting lower residual trapping. The dashed line (1:1) indicates complete trapping (100% retention). The increasing separation from the 1:1 line reflects growing hysteresis and trapping with cycling.

illustrates a clear hysteretic trend, with both the initial saturation ($S_{H_2,i}$) and the corresponding residual saturation ($S_{H_2,r}$) evolving across cycles. In the first cycle, $S_{H_2,i}$ peaks at 0.847 with a corresponding $S_{H_2,r}$ of 0.600. Both values increase in the second cycle, likely due to fluid redistribution and evolving pore connectivity that impact gas mobility. Residual saturation continues to rise slightly over successive cycles, reaching a maximum of 0.64 by the fifth cycle. This increasing trend suggests progressive gas phase fragmentation, where snap-off events and gas discontinuity become more pronounced due to capillary trapping mechanisms. Beyond the fifth cycle, however, residual saturation begins to decline slightly (sixth and seventh cycles), eventually stabilizing with differences falling below 0.02. This convergence indicates the onset of hysteresis equilibrium, a state where further cycling no longer significantly alters the phase distribution or residual saturation. The trend observed here is consistent with previous studies in sedimentary rocks, where similar plateau-like behaviour has been reported. For example, Yang et al. (2025) observed saturation stabilization after four cycles, while Wang et al. (2023a) reported comparable convergence by the fifth cycle. These results reinforce the interpretation by Zhang et al. (2023), who demonstrated that snap-off and increasing non-wetting phase discontinuity during cyclic imbibition gradually fill available capillary traps, leading to saturation asymptotes. This trend, where residual trapping increases over successive cycles before plateauing, likely results from the cumulative effects of snap-off and enhanced gas phase fragmentation in angular throat geometries. As observed earlier, basalt pores exhibit high aspect ratios and low shape factors, which support wetting layer swelling and capillary pinch-off. Such geometric features promote the breakup and entrapment of H_2, contributing to the observed hysteresis and increasing residual saturation over subsequent cycles. Thus, the approach to hysteresis equilibrium reflects a balance between pore-scale trapping capacity and the redistribution dynamics of the invading brine.

Our findings have far-reaching implications for full-field UHS design, as the high residual gas saturation suggests enhanced storage security but at the cost of reduced deliverability. This trade-off arises because a residual saturation exceeding 60% implies that a significant portion of the injected H_2 becomes immobile due to capillary trapping. While this immobile fraction helps maintain reservoir pressure and prevents brine up-coning (Nazari et al., 2024), it also limits the volume of hydrogen available for withdrawal. A field-scale simulation with similar residual values show that the working-to-total gas ratio drops to roughly one-quarter of the in-place volume when hysteresis is accounted for (Lysyy et al., 2023). In contrast, models that neglect capillary trapping tend to over-predict deliverability by 30%–40% and the recoverable volume by up to 85% (Lysyy et al., 2023). The observed hysteresis equilibrium indicates that pore-scale trapping and release dynamics become more predictable after an initial conditioning phase. From an operational standpoint, this could guide the optimization of cycling frequency by allowing operators to avoid inefficiencies during early cycles when saturation changes are more variable. Intentional pre-conditioning (e.g., performing a few cycles using cushion gas such as CH_4 and N_2) could be employed to reach hysteresis equilibrium before commencing full-capacity cyclic operations, thereby optimizing long-term performance and minimizing hydrogen losses due to variable trapping behaviour. Furthermore, the attainment of hysteresis equilibrium provides insight into cushion gas requirements, suggesting that once equilibrium is achieved, a consistent and potentially lower volume of cushion gas may suffice to maintain performance. Incorporating

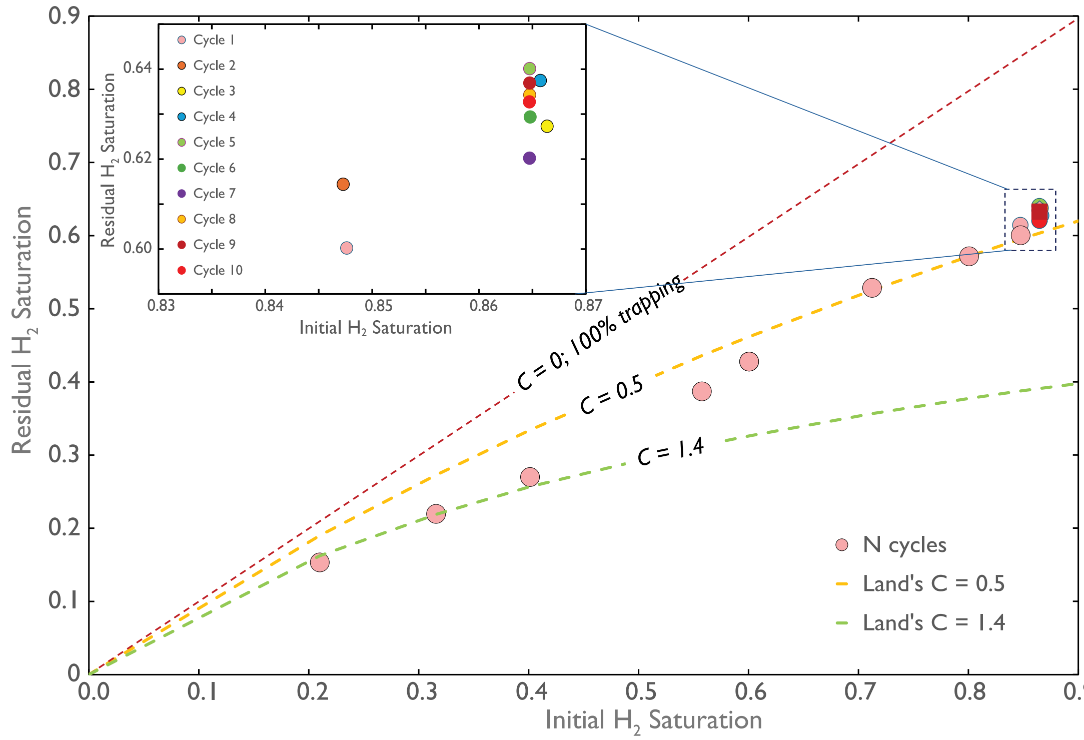

Fig. 13. Initial H_2 saturation ($S_{H_2,i}$) at the end of the nth drainage cycle versus residual (trapped) H_2 saturation ($S_{H_2,r}$) at the end of the corresponding nth imbibition cycle. Cycle number (n) is denoted by colour.

these insights into UHS protocols could enhance both the efficiency and reliability of cyclic hydrogen injection and withdrawal operations.

3.6. Effect of wettability alteration on hydrogen saturation and withdrawal efficiency

Wettability in subsurface environments is inherently dynamic and can shift due to geochemical reactions, temperature variations, or microbial activity, particularly within reactive lithologies such as basalt (Aftab et al., 2024; Al-Yaseri et al., 2023; Boon et al., 2024a). These shifts can strongly influence fluid behaviour at the pore scale, modifying both the extent of hydrogen trapping and the efficiency of hydrogen recovery. Building on strongly water-wet baseline discussed previously, we incorporate simulations of weakly water-wet conditions to examine how alterations in wettability influence capillary pressure, relative permeability, and phase displacement dynamics. As shown in Fig. 14, transitioning from strongly to weakly water-wet states yields significant changes in relative permeability and capillary pressure behaviour across both drainage and imbibition.

The weakly water-wet conditions results in a reduction in capillary pressure, leading to higher gas saturations at lower pressure thresholds. This trend, which becomes more pronounced at low to intermediate pressure levels during imbibition, corresponds with increases in the contact angle (θ) and aligns with findings reported by Li et al. (2024). Strong water-wet conditions increase the affinity between the rock surface and the brine, promoting the development of stable wetting films (Wang et al., 2024). These films foster snap-off events, particularly within angular throats, reducing gas mobility and connectivity and subsequently lowering relative permeability, as observed in Fig. 14b. By contrast, weakly water-wet conditions minimizes snap-off, enabling greater gas mobility and lowering residual hydrogen saturation from 0.60 to 0.55, which translates into enhanced recoverability. These effects are further visualized in Fig. S7, which presents 3D renderings of the pore network under both wettability regimes. During drainage, gas invasion patterns appear broadly similar; however, during imbibition, strongly water-wet conditions result in more localized trapping and disconnected hydrogen clusters. In contrast, weakly water-wet conditions promote more extensive brine invasion and yield fewer isolated gas pockets, indicating enhanced displacement efficiency and a reduction in residual trapping.

Comparable results were reported by Al-Yaseri et al. (2023) and Liu et al. (2023), who demonstrated that surface chemical alterations promoting less water-wet states significantly reduced residual hydrogen saturation (from 23% to 11%). Such observations indicate that optimizing wettability conditions away from strong water-wetness can be beneficial to minimize capillary trapping and enhance gas recovery. Similarly, Aftab et al. (2024) reported that sulphate-reducing bacteria modified surface wettability in basalt from strongly to weakly water-wet, significantly reducing capillary pressure by 65%. However, such microbial activity may also introduce challenges, including potential hydrogen loss through biological consumption or contamination by metabolic byproducts such as hydrogen sulfide (Aftab et al., 2024; Boon et al., 2024a,b). Despite these parallels, our model does not consider time-dependent, wettability-altering processes as discussed in Section 2.2. For example, pore clogging from microbial biomass could increase irreducible brine saturation, while non-uniform wettability distribution might enhance snap-off events or promote disconnected gas clusters. These evolving conditions can either amplify or suppress observed hysteresis over successive cycles, depending on the extent and evolution of microbial activity. Such complexities underscore the need for future studies that couple biofilm growth, decay, and related microbial interactions with multiphase flow and transport models to better represent field-scale conditions in UHS operations.

To better understand the pore-scale mechanisms (Fig. 2) responsible for these wettability-dependent trends, we conducted detailed analysis of the frequency and distribution of displacement events within the pore network, as presented in Fig. 15. The overall higher median of snap-off events (Fig. 15a and c) under strongly water-wet conditions, compared to the weakly water-wet case, highlights the dominance of snap-off mechanisms, particularly within angular throats and high aspect-ratio pore bodies, leading to fragmentation of the non-wetting

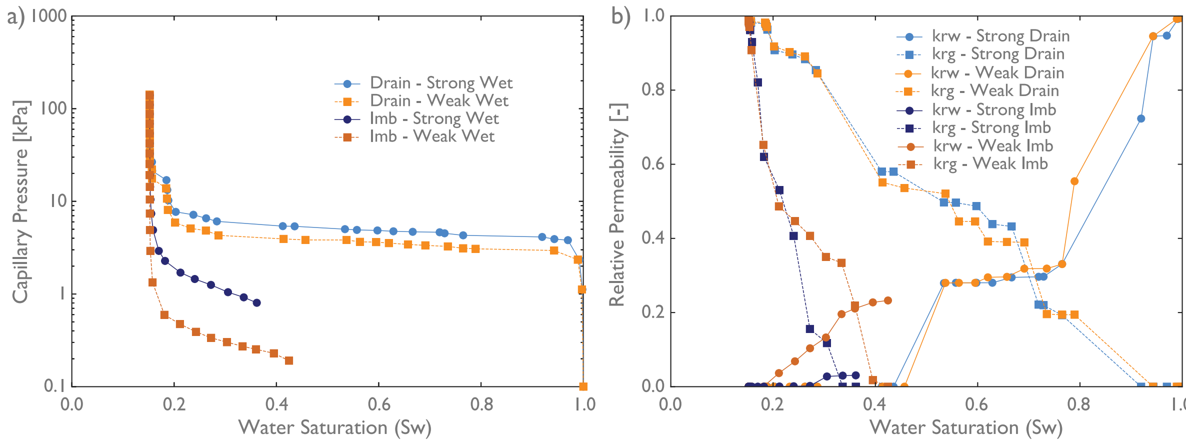

Fig. 14. Influence of wettability on capillary pressure and relative permeability during hydrogen–brine displacement in basalt. (a) Capillary pressure–saturation relationships for both drainage and imbibition cycles show that weakly water-wet conditions significantly reduce capillary entry pressure and overall capillary resistance, enabling earlier and more uniform hydrogen invasion and brine displacement. The observed reduction in hysteresis under weakly water-wet conditions highlights enhanced accessibility and connectivity for the invading phase. (b) Relative permeability curves demonstrate that weakly water-wet systems improve hydrogen phase mobility across the saturation range, while also supporting more efficient brine re-invasion during imbibition. In contrast, strongly water-wet conditions lead to more pronounced hysteresis and increased residual trapping of the non-wetting hydrogen phase. These trends emphasize the key role of wettability in governing pore-scale displacement mechanisms, connectivity, and recovery efficiency.

hydrogen phase into disconnected ganglia. While the median frequency of snap-off is slightly lower in pore bodies (Fig. 15a), the lower capillary entry pressure associated with strong water-wet conditions allows hydrogen to access smaller pores more readily during drainage. This increased accessibility raises the likelihood of snap-off events during imbibition, although our analysis shows only a marginal difference in the snap-off frequency compared to water-wet conditions. Importantly, the magnitude of difference in snap-off occurrence is more pronounced within throats than pores, reinforcing the conclusion that snap-off is more dominant in strongly water-wet systems. This observation aligns with the pore-scale displacement behaviours and geometric trapping tendencies discussed earlier, particularly the role of enhanced film swelling under strong water-wet affinity conditions.

Similar trends are observed in the higher median frequencies for piston-like displacements (Fig. 15b, d) and pore body filling events dominated by I_2 and I_3 mechanisms (Fig. 15e, f), which demonstrate a notable increase under weakly water-wet conditions relative to strongly water-wet cases. These displacement mechanisms facilitate smoother, more continuous brine invasion, minimize gas discontinuities, and improve hydrogen phase connectivity, contributing to lower residual saturation. This behaviour aligns with the findings of Peng et al. (2025), who employed volume-of-fluid simulations to show that weakly water-wet systems inhibit wetting layer formation, reduce solid-fluid interactions, and lower capillary entry pressures. In general, these observations suggest that wettability alteration not only governs local pore-scale invasion dynamics but also shifts the displacement regime from snap-off dominated to piston-like behaviour, thereby minimizing disconnected hydrogen clusters and maximizing recoverable gas volumes.

We assessed the evolution of hydrogen residual saturation and recovery efficiency across multiple drainage–imbibition cycles to quantify the impact of wettability on hydrogen recovery. Quantitatively, the differences in wettability translate into measurable impacts on recovery efficiency, shown in Fig. 16, which is defined as the proportion of recoverable hydrogen saturation compared to the amount of gas injected following drainage. Simulation results (Fig. 16) show a consistent increase in residual gas saturation across cycles, particularly under strongly water-wet conditions. From the first to the fifth cycle, a slight but systematic rise is observed, which may be attributed to increased hydrogen snap-off and cluster breakup and subsequent disconnected hydrogen ganglia formation. These disconnected gas clusters accumulate as repeated cycles redistribute hydrogen from smaller to larger pores, where trapping is more persistent. Wettability alteration from strongly to weakly water-wet states results in a measurable improvement in hydrogen recovery. As shown in Fig. 16, recovery efficiency improves from ~29% under strongly water-wet conditions, where residual saturation stabilizes around 62%, to about ~35% under weakly water-wet conditions, where residual saturation decreases to 55%. This ~6% enhancement highlights the substantial role that wettability plays in governing displacement efficiency and recoverable hydrogen volumes. These results are consistent with prior findings of Al-Yaseri et al. (2023), who reported a 6% net gain in hydrogen recovery from 20.7% to 26.7% as a result of humic acid treatment of a sandstone reservoir rock. Similarly, Bahrami et al. (2024) reported an increase in trapped gas saturation from 38.7% to 42.8% with successive cycles and demonstrated that residual saturation correlates with the initial gas saturation in each cycle.

Overall, our pore-scale modelling reveals that wettability alteration towards less water-wet conditions reduces capillary trapping, delays gas disconnection, and facilitates more effective brine displacement, ultimately enhancing hydrogen recovery during cyclic injection and withdrawal. These shifts in invasion dynamics and trapping behaviour underscore the role of surface wettability as a key control on subsurface gas retention and mobilization. When considered alongside the geometric constraints previously discussed, such as high aspect ratios and low shape factors, wettability emerges as a complementary, second-order factor influencing recovery efficiency in basaltic pore networks. Importantly, because wettability can be modified in situ through chemical or biological treatments, it presents a potential lever for engineering UHS improvements in basaltic reservoirs. Strategic tailoring of wettability, alongside geometric entrapment insights, could enable more efficient gas withdrawal while ensuring adequate trapping for long-term storage security.

4. Conclusions

Basalt formations are gaining increasing attention as candidates for underground hydrogen storage (UHS) due to their widespread availability and inherent storage potential. Yet, understanding how hydrogen is trapped or mobilized under cyclic injection and withdrawal and wettability variations remains limited, particularly at the pore scale. This study provides a comprehensive analysis of H_2-brine interactions within basalt rock utilizing pore scale modelling strategy, with a focus on how wettability, pore geometry, and cyclic operation influence trapping behaviour, residual saturation, and recovery efficiency.

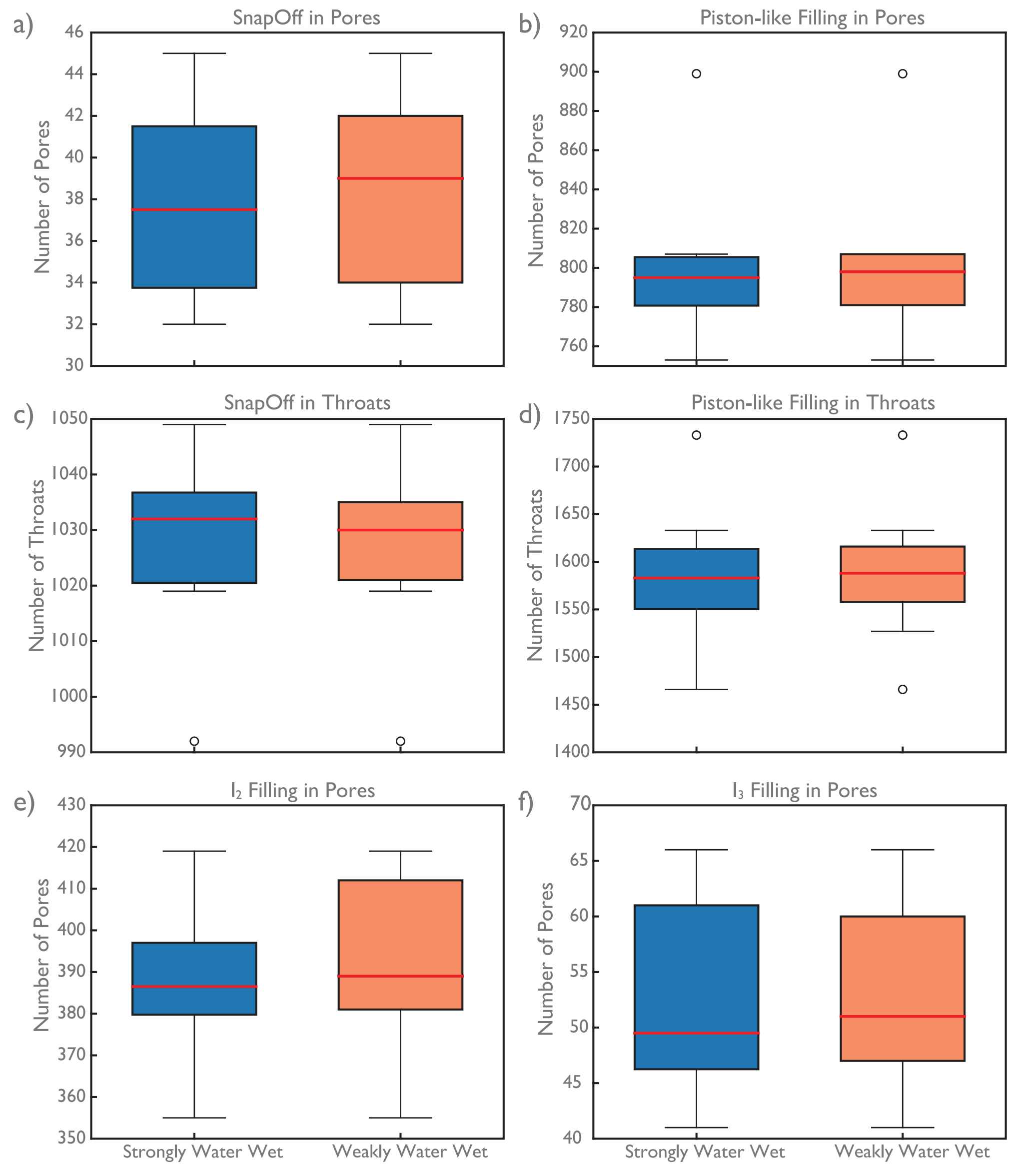

Fig. 15. Box plots showing the distribution of pore and throat filling events controlled by snap-off and piston-like mechanisms under strongly and weakly water-wet conditions. Pore-filling events through (a) snap-off and (b) piston-like mechanisms. Throat-filling events through (c) snap-off and (d) piston-like mechanisms (e) I_2 and (f) I_3 type pore body filling mechanism.

Our findings reveal that during drainage, H_2 preferentially invades larger pores due to their lower entry capillary pressures. However, these same pores become preferential zones for residual trapping during imbibition, due to geometric constraints, including low shape factors, high aspect ratios, and restricted throat connectivity which are characteristic of our basaltic formations. The interplay between pore-throat geometry and wetting layer dynamics significantly controls hydrogen retention, with narrow or angular throats serving as critical locations for snap-off events and gas phase fragmentation. Through repeated drainage-imbibition cycles, the residual hydrogen saturation increased modestly, stabilizing around 61%–62% after the fifth cycle. This behaviour suggests a progressive evolution towards hysteresis equilibrium, where additional cycling no longer significantly alters the trapped phase distribution. This pattern is indicative of cumulative snap-off and gas phase fragmentation, which progressively fill available capillary traps until saturation is achieved. The hysteresis trends align with known trapping mechanisms in porous media and affirm basalt's capacity to retain H_2 under repeated injection and withdrawal cycles. While these trapping characteristics provide essential storage security, they create a direct trade-off with recovery efficiency by limiting phase connectivity and reducing gas mobility. The extent of this trade-off depends critically on wettability conditions and initial saturation levels. Quantitatively, the Land's trapping coefficients derived for basalt (0.5–1.4) fall below those typical of carbonates but align closely with sandstone values, indicating moderate to strong trapping capacity that intensifies under strongly water-wet conditions.

The enhanced phase trapping and lower recovery efficiency observed under strongly water-wet conditions stem from elevated capillary trapping forces that restrict hydrogen mobility. On the other hand, the systematic shift towards lower capillary pressures and higher relative permeabilities in weakly water-wet systems demonstrates the potential of wettability modification to improve both storage performance and recovery efficiency. Consequently, understanding and controlling wettability becomes critical for managing the fundamental trade-off between storage security and H_2 recoverability, particularly in ensuring efficient extraction during periods of peak energy demand. While our simulations assume quasi-static, capillary-dominated flow, which is appropriate under low-rate injection conditions, field-scale operations span a wider range of flow regimes. Sharp permeability contrasts can exacerbate channelling in high-permeability zones and

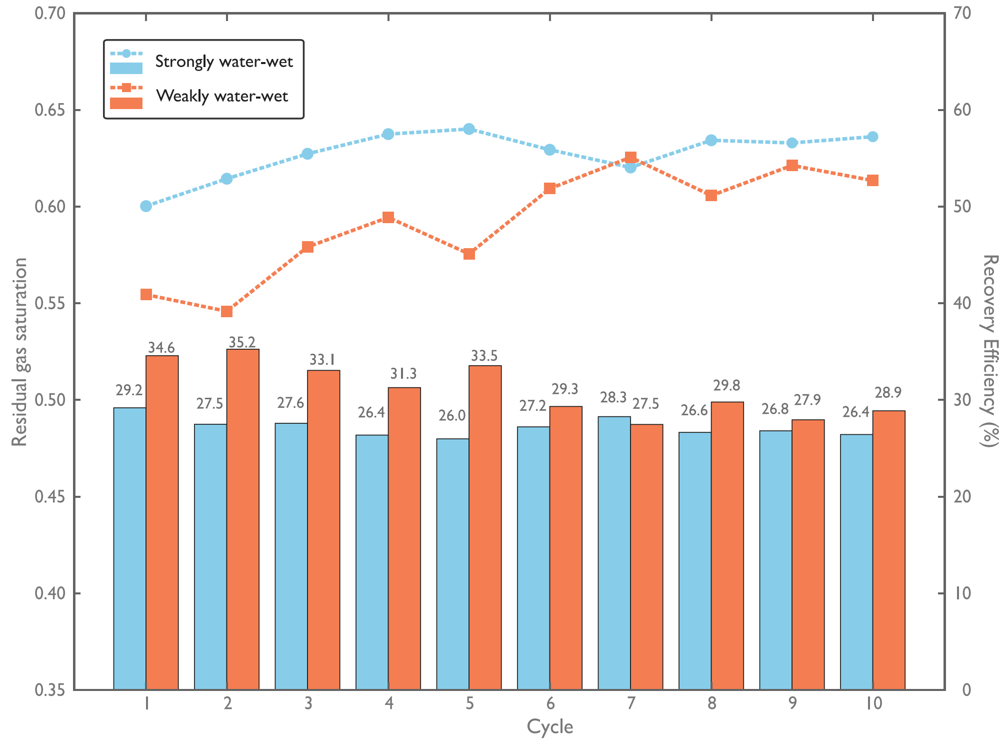

Fig. 16. Effect of wettability variation on trapped H_2 saturation, recoverable hydrogen, and withdrawal efficiency. Dotted lines indicate the residual gas saturation, while bar charts represent the corresponding recovery efficiency.

lead to higher capillary numbers. Under such conditions, the displacement front may become destabilized, triggering viscous fingering that bypasses portions of the pore volume. As a result, higher brine rates may remobilize previously trapped hydrogen during withdrawal by overcoming capillary forces that held them immobile, leading to lower residual saturation than those predicted under quasi-static assumptions. These dynamic effects underscore the need to complement pore-scale, quasi-static predictions with considerations of viscous forces and reservoir heterogeneity when forecasting deliverability at scale. Notwithstanding, the findings in this study advance the understanding of hydrogen storage efficiency in basaltic lithologies and support improved reservoir design and operational optimization.

CRediT authorship contribution statement

Enoch Ibitogbe: Writing – original draft, Methodology, Formal analysis, Data curation, Visualization. **Adedapo N. Awolayo:** Writing – review & editing, Conceptualization, Visualization, Validation, Resources, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.advwatres.2025.105068.

Data availability

Data will be made available on request.

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